

cut-condition.cc

Lambda cut study from $p + \pi^-$

runs 02447-02460, n=12

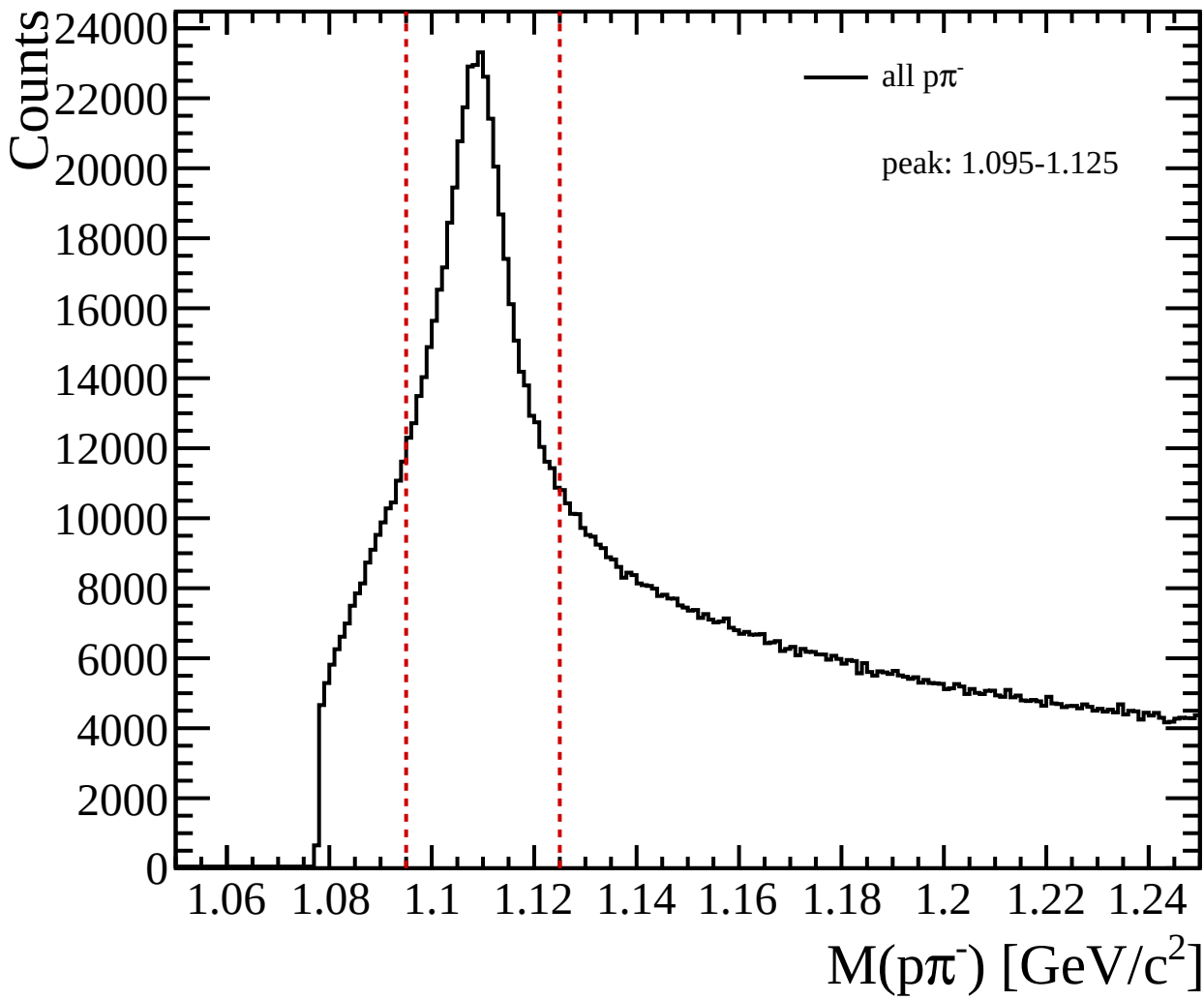
comparison: all $p\pi^-$, Lambda peak, and tight quality cut

peak window: $1.095 < M(p\pi^-) < 1.125 \text{ GeV}/c^2$

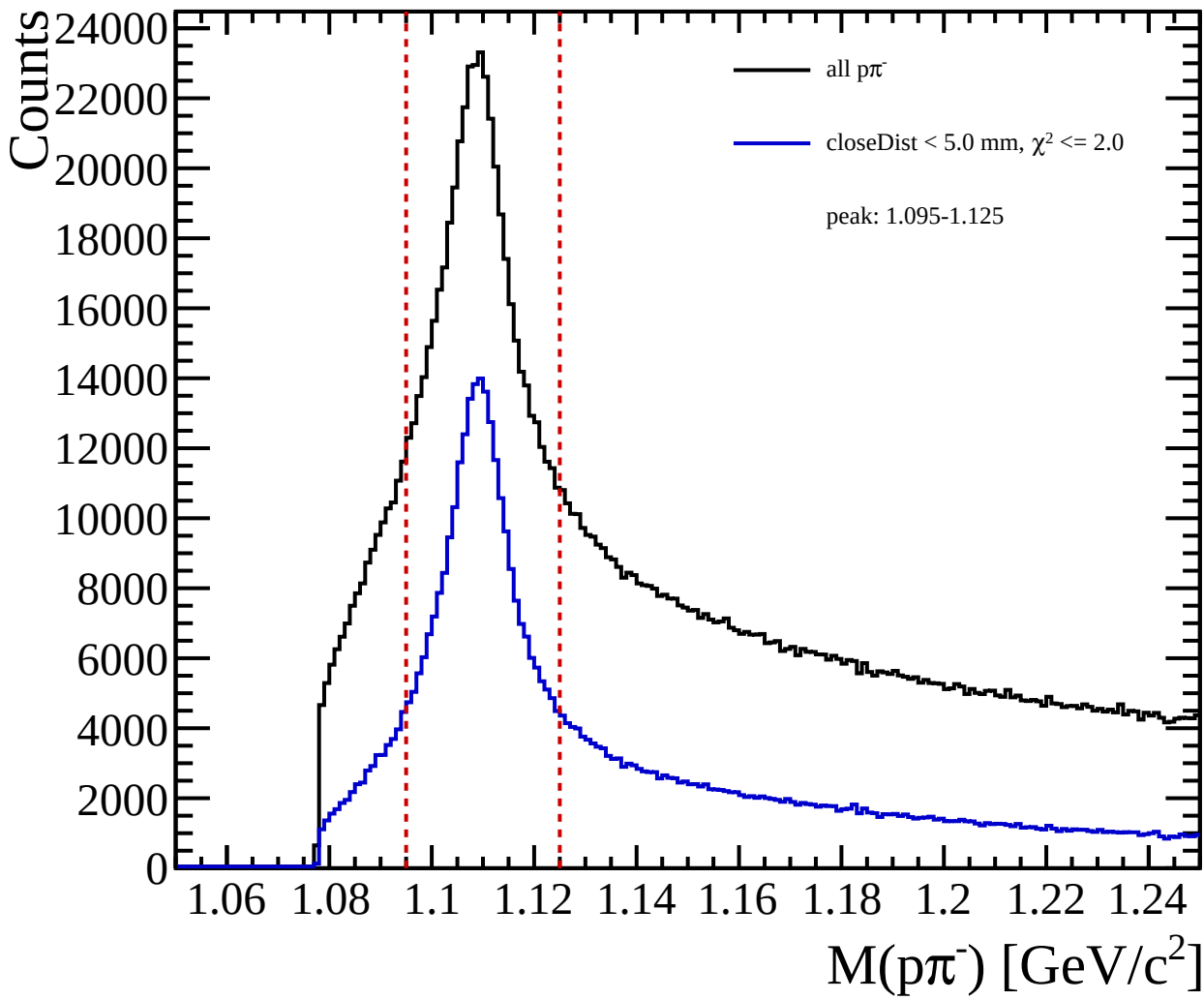
tight cut: $\text{closeDist} < 5.0 \text{ mm}$, $\chi_p^2 \leq 2.0$, $\chi_\pi^2 \leq 2.0$

Generated at: 2026-07-02 15:35:46

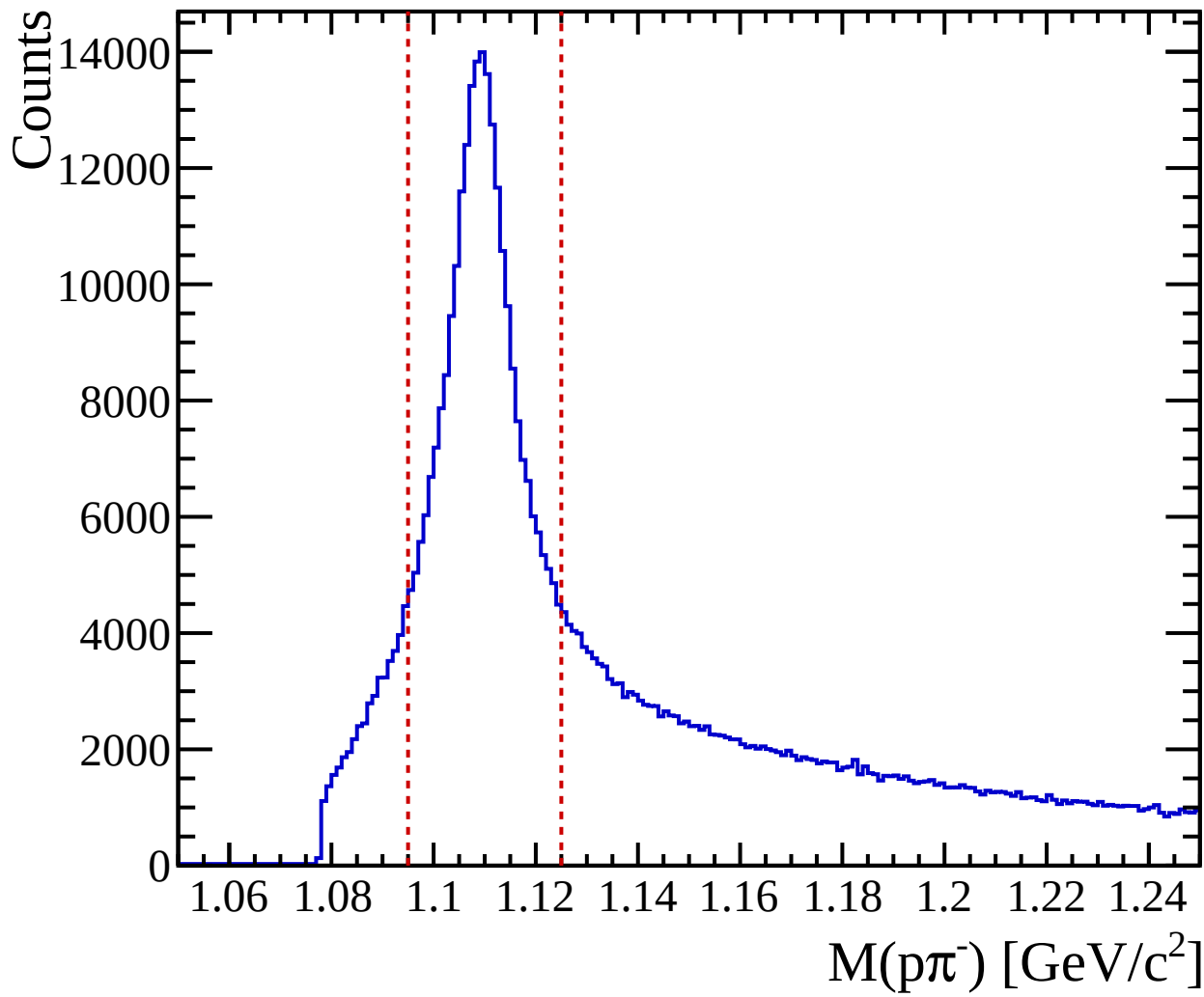
$p\pi^-$ invariant mass



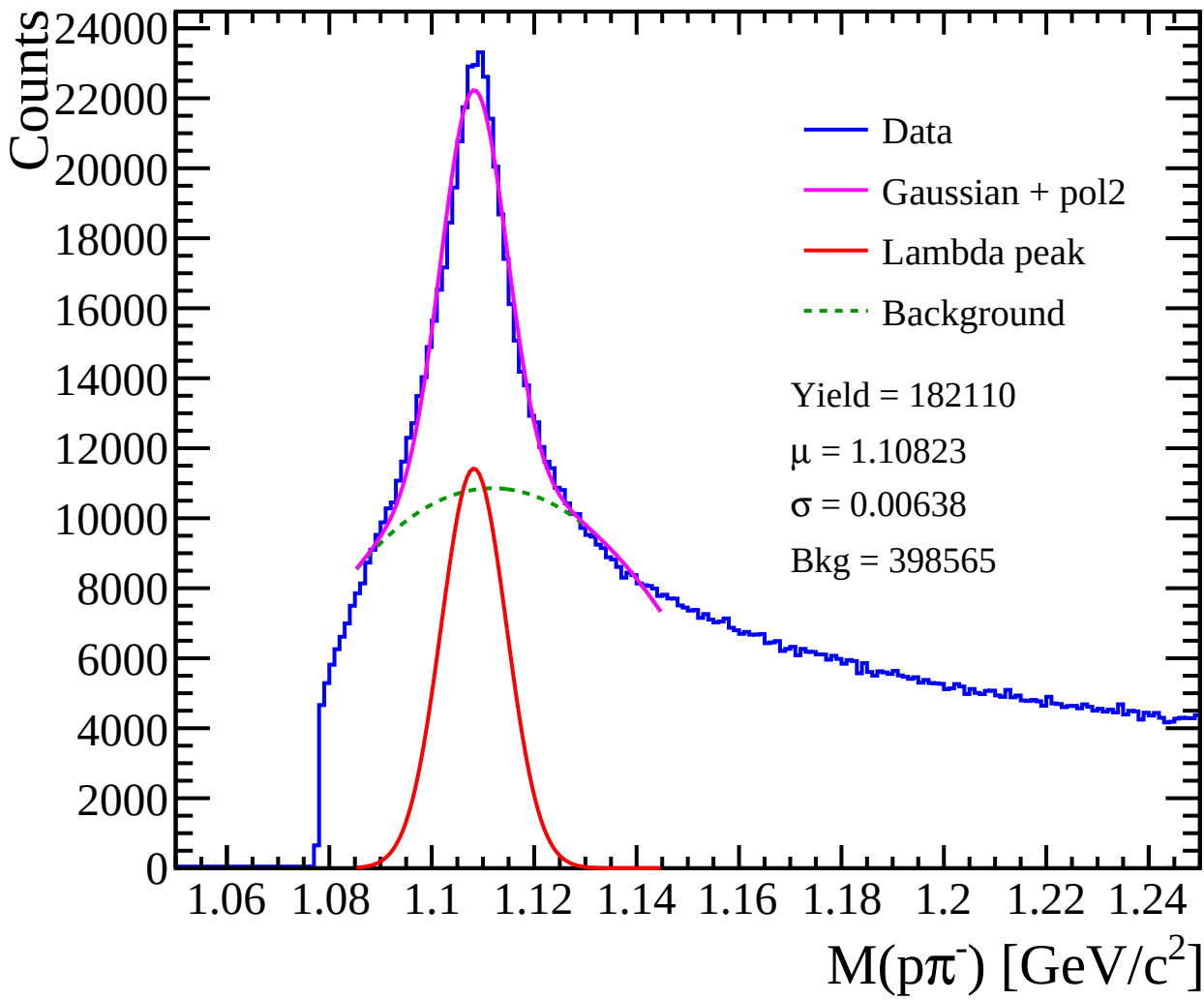
$p\pi^-$ invariant mass



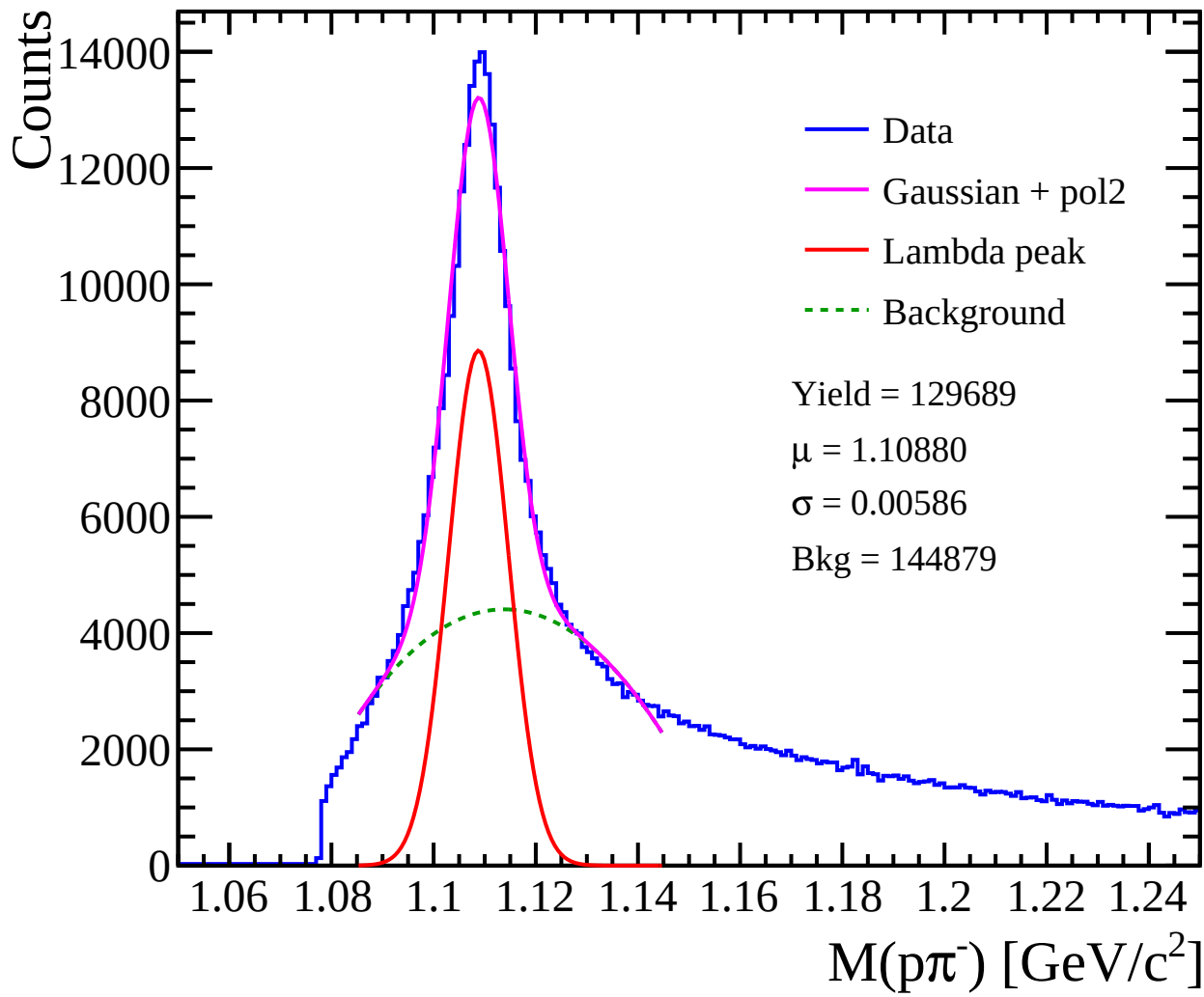
$p\pi^-$ invariant mass, tight cut



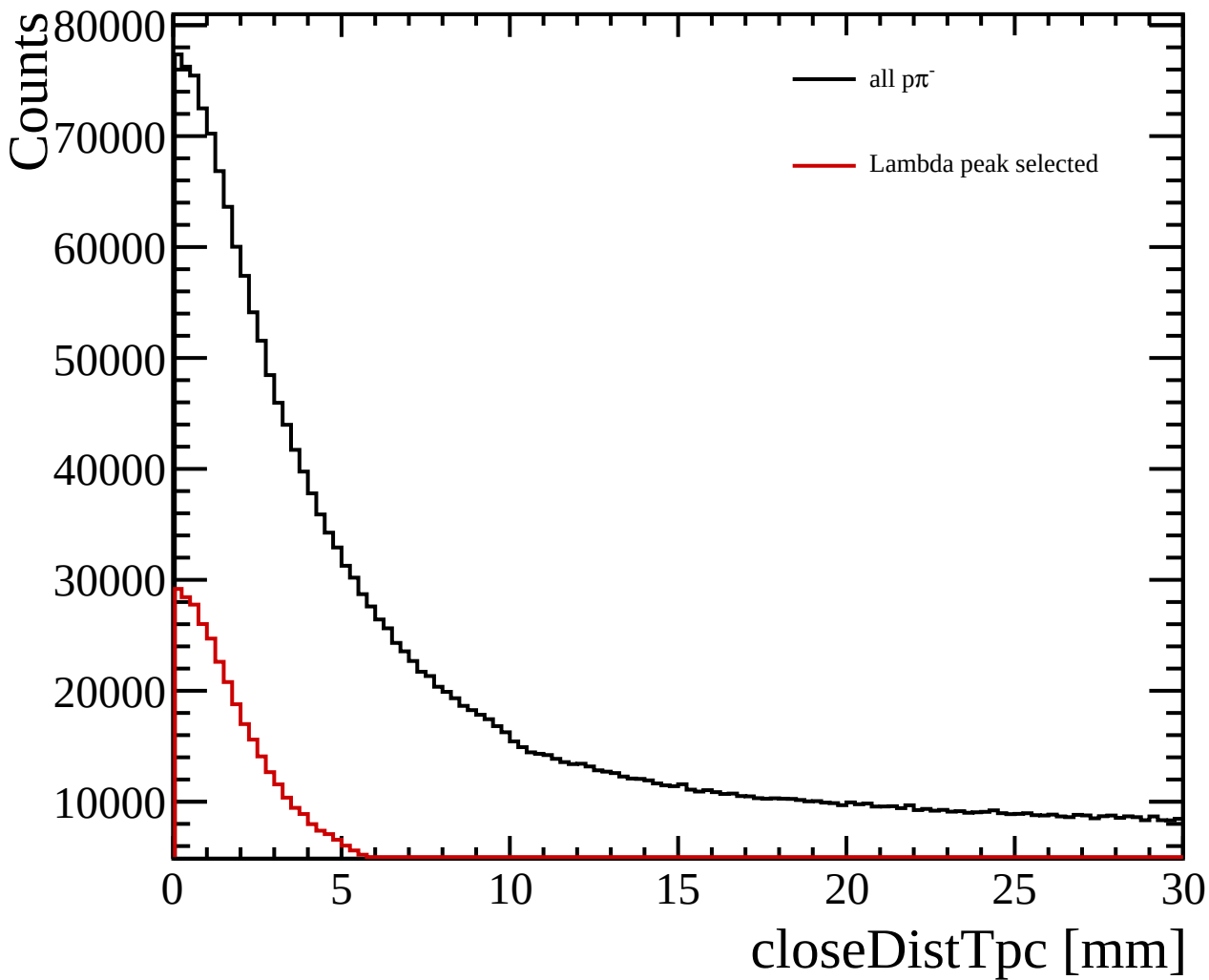
$p\pi^-$ invariant mass



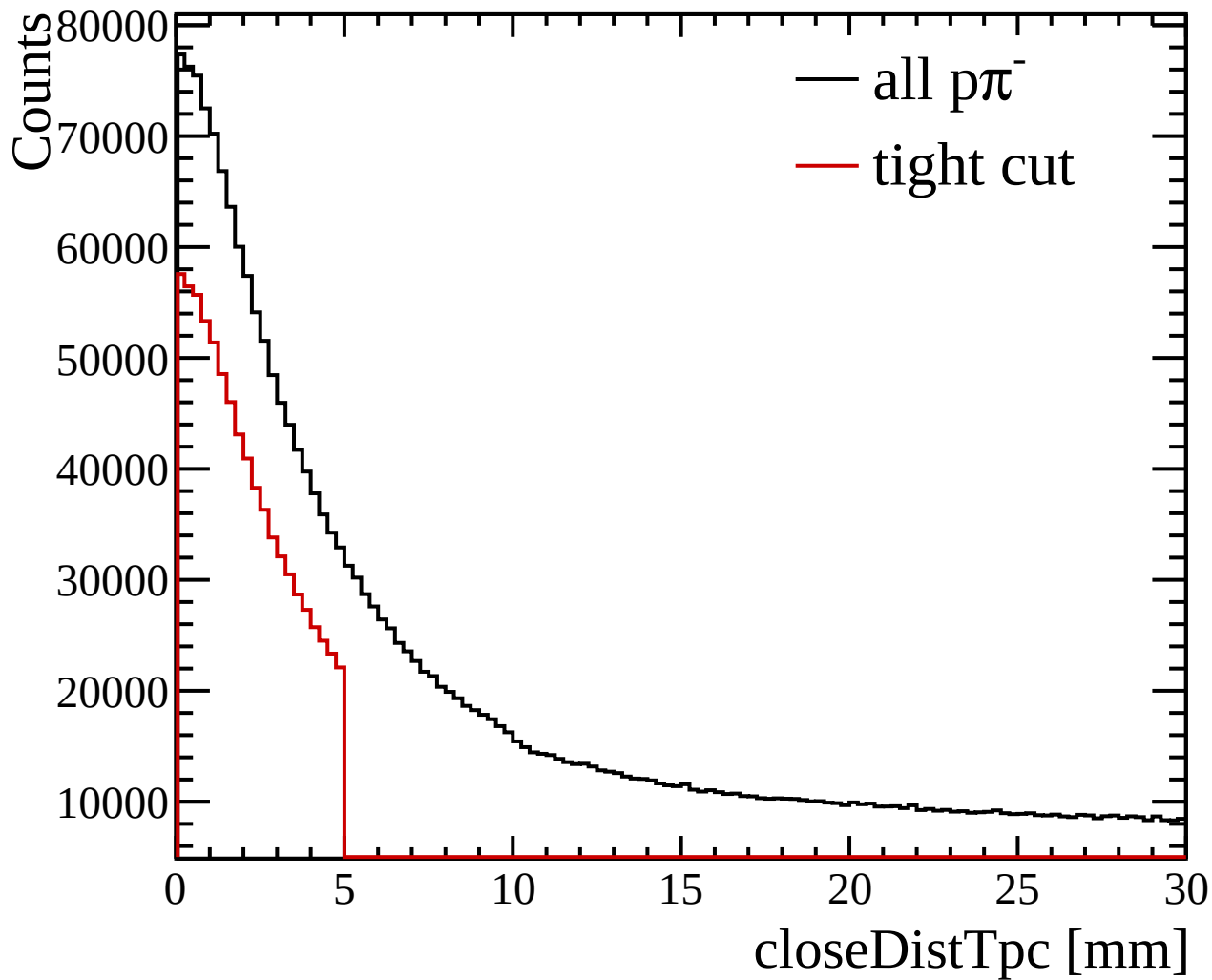
$p\pi^-$ invariant mass, tight cut



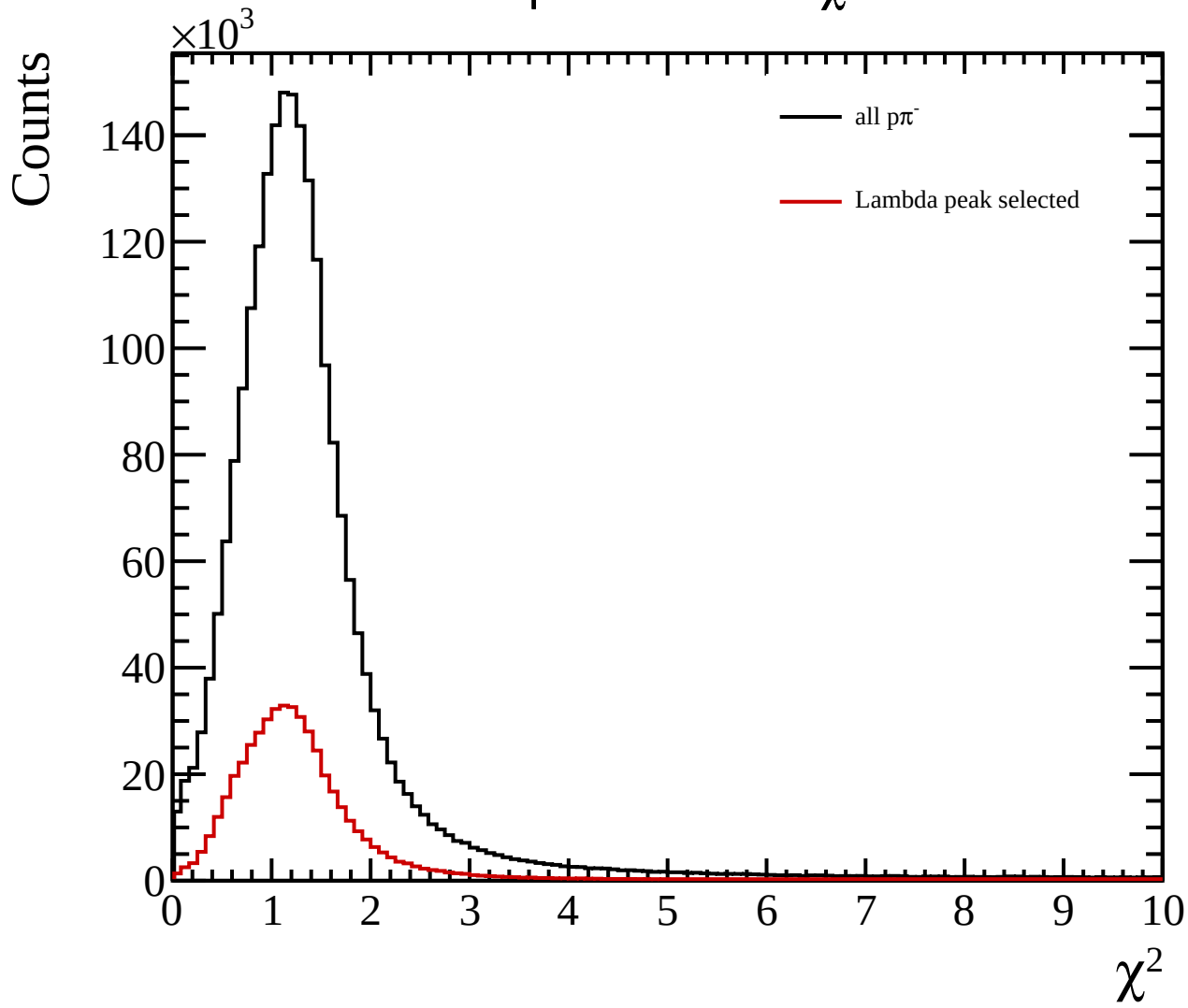
$p\pi^-$ close distance



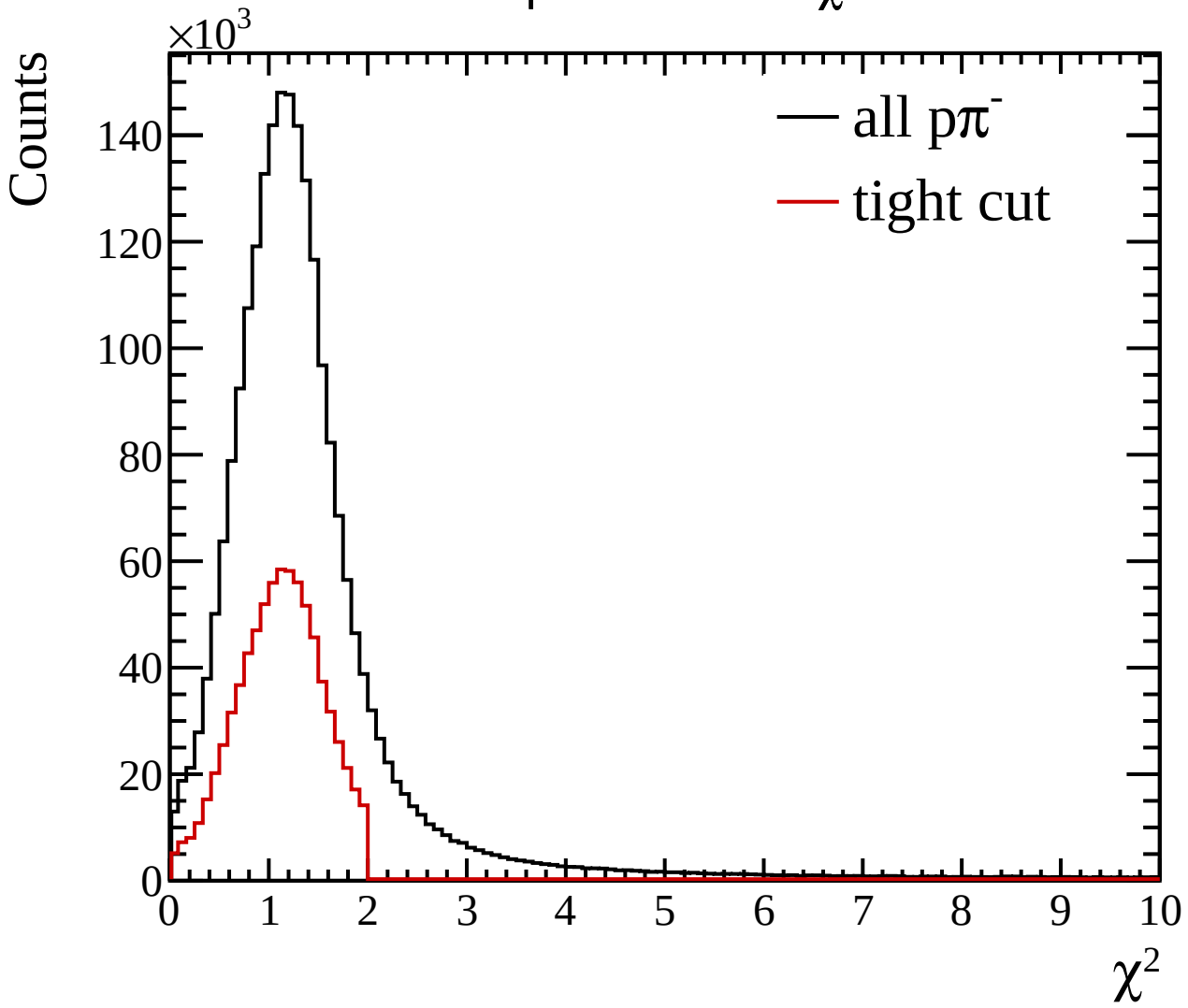
$p\pi^-$ close distance



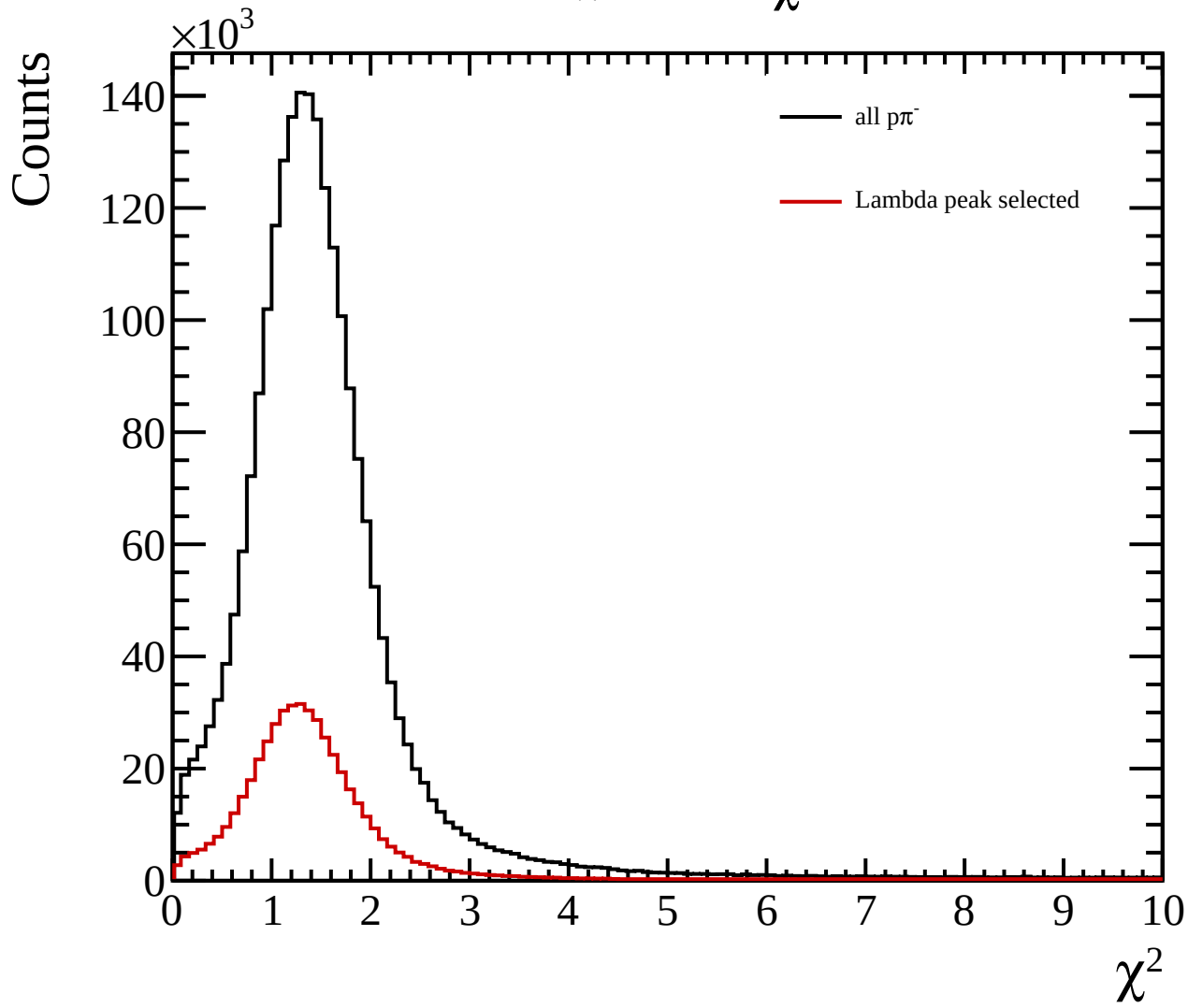
proton track χ^2



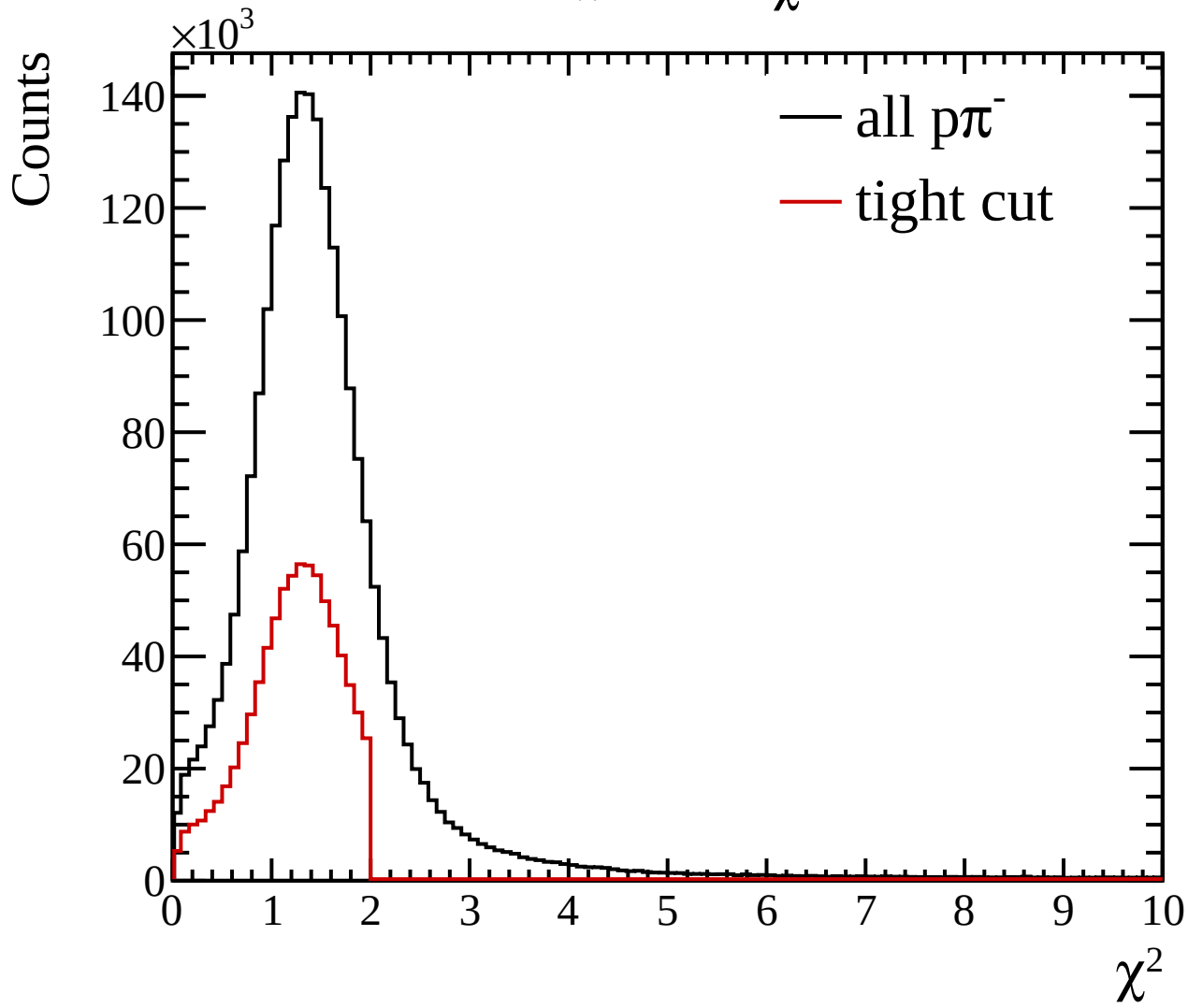
proton track χ^2



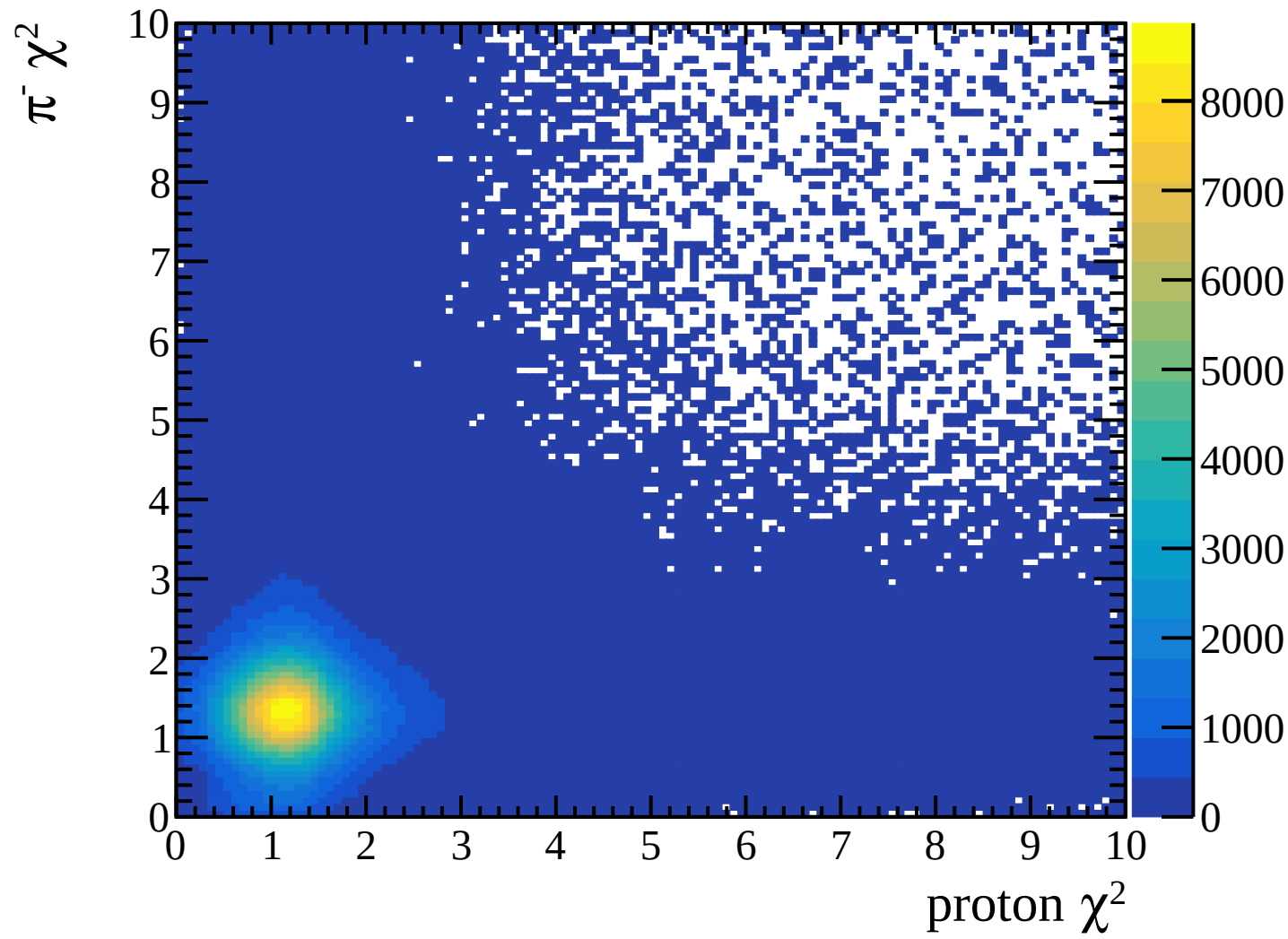
π^- track χ^2



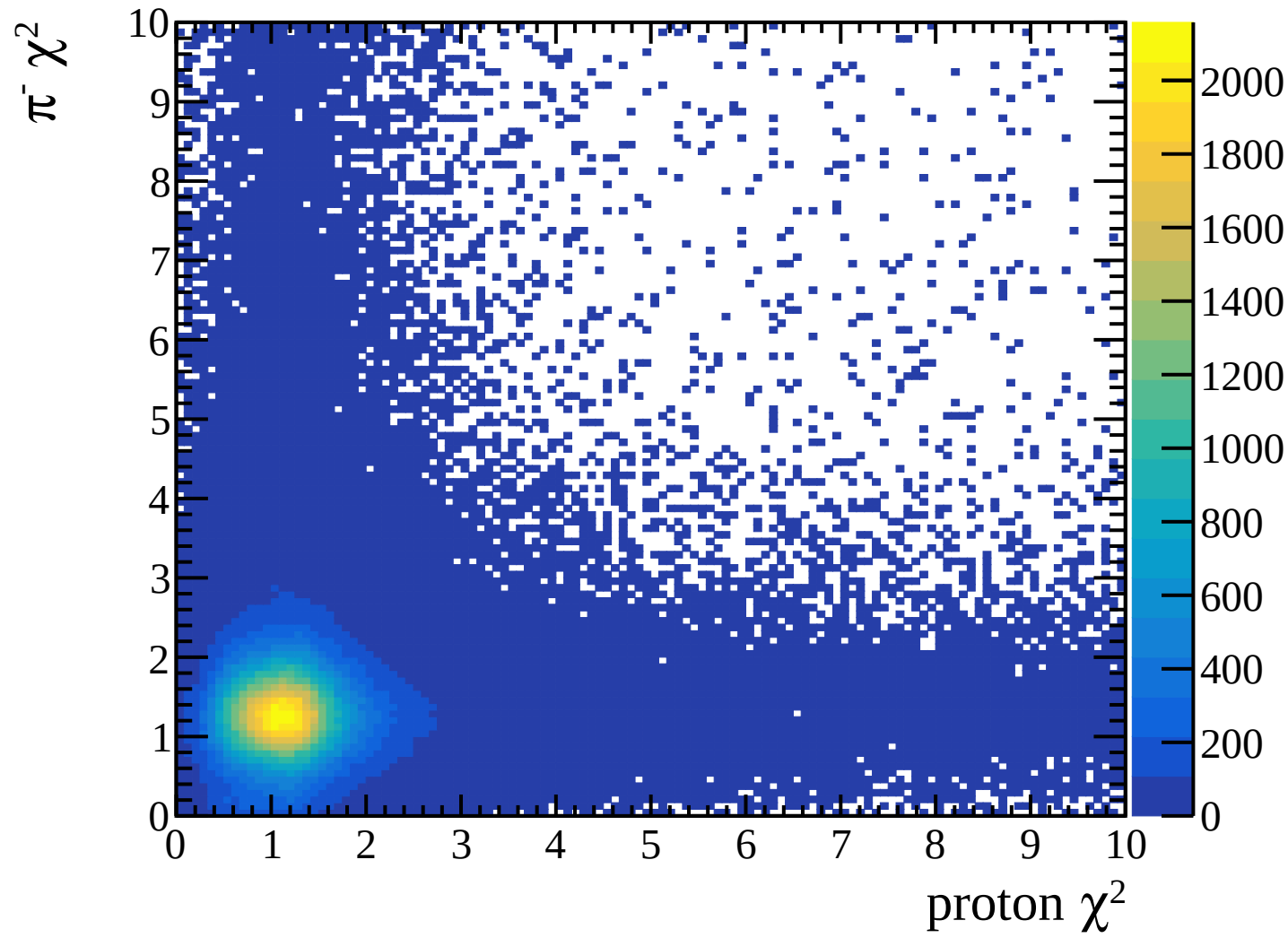
π^- track χ^2



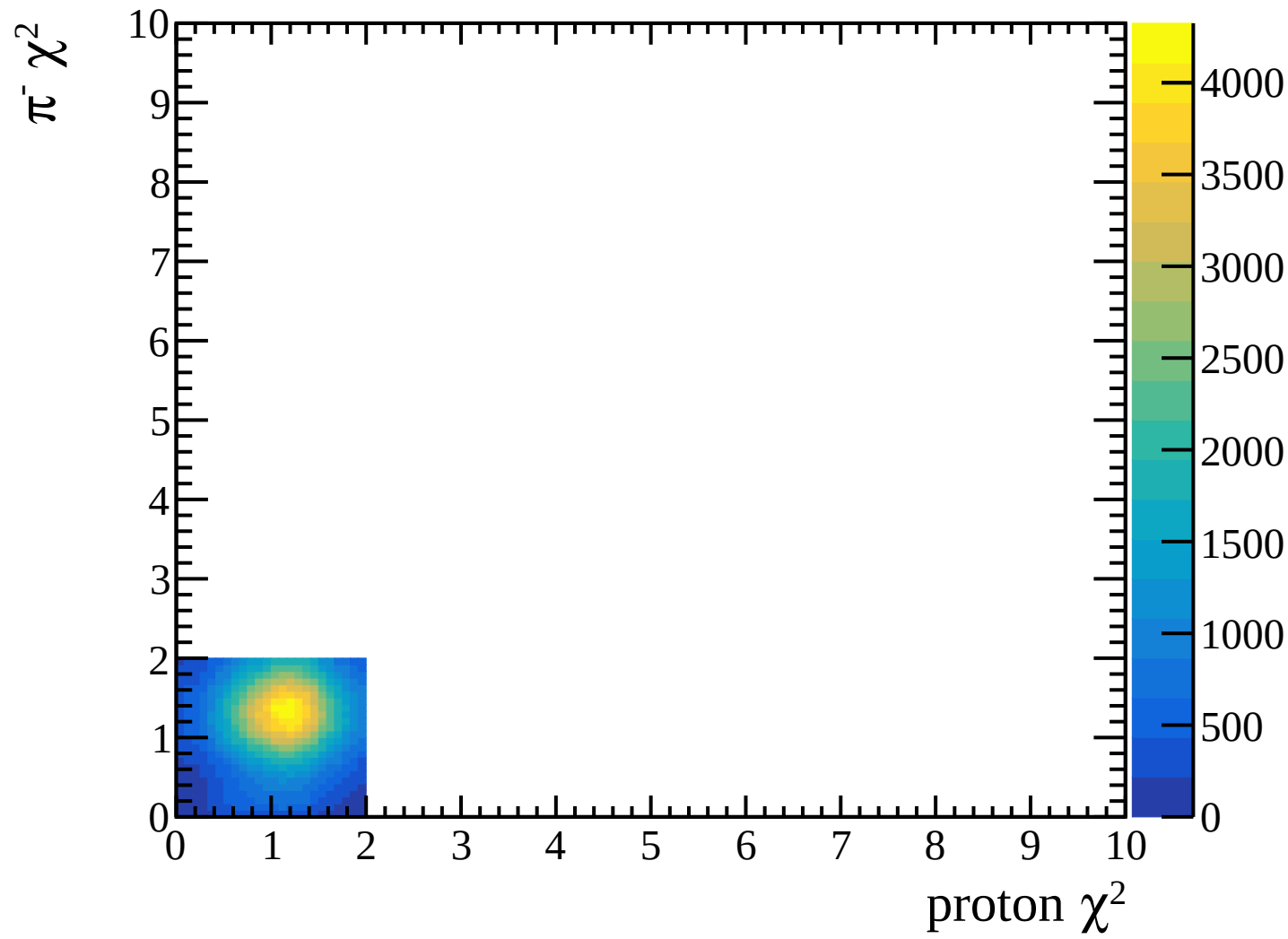
track χ^2 , all $p\pi^-$



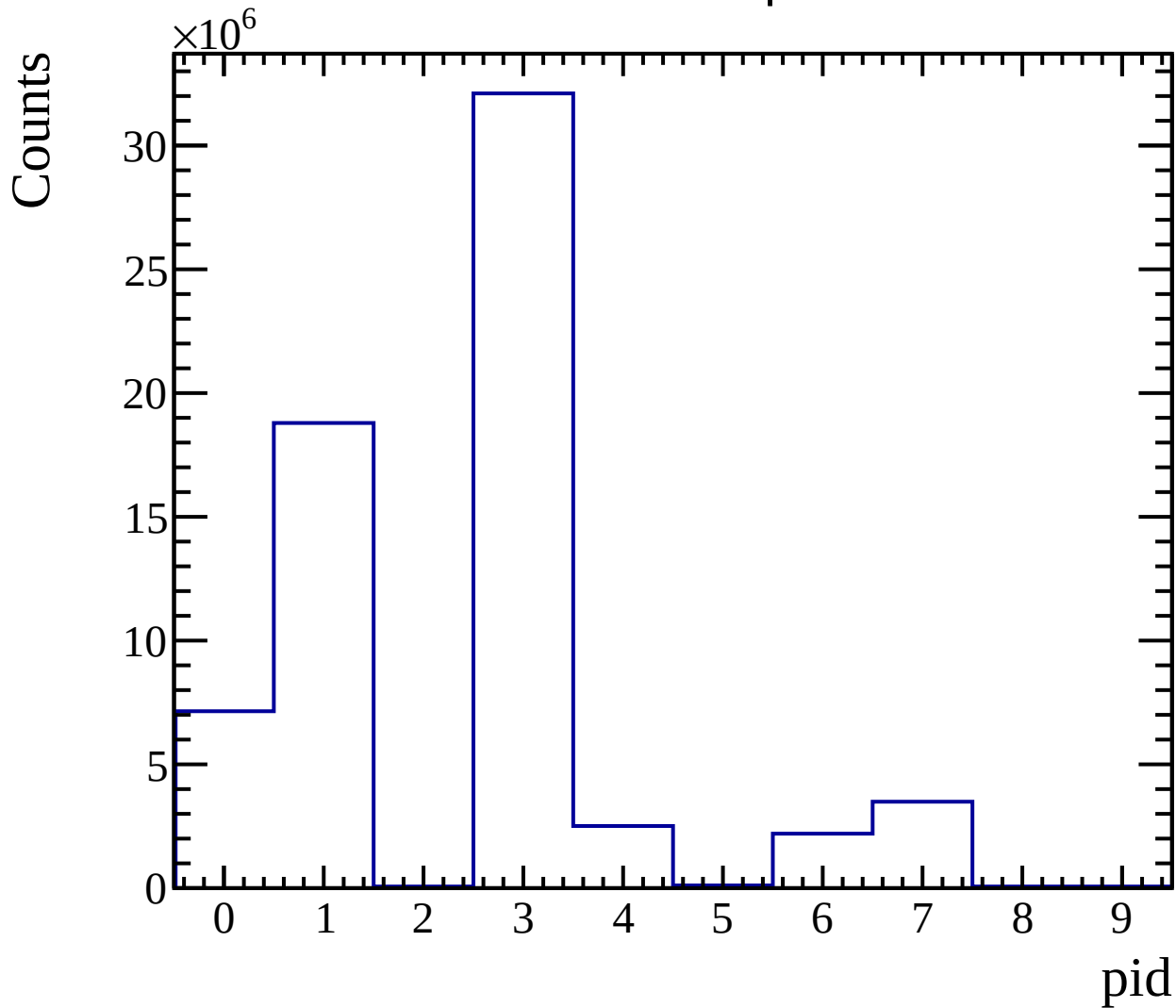
track χ^2 , Lambda peak



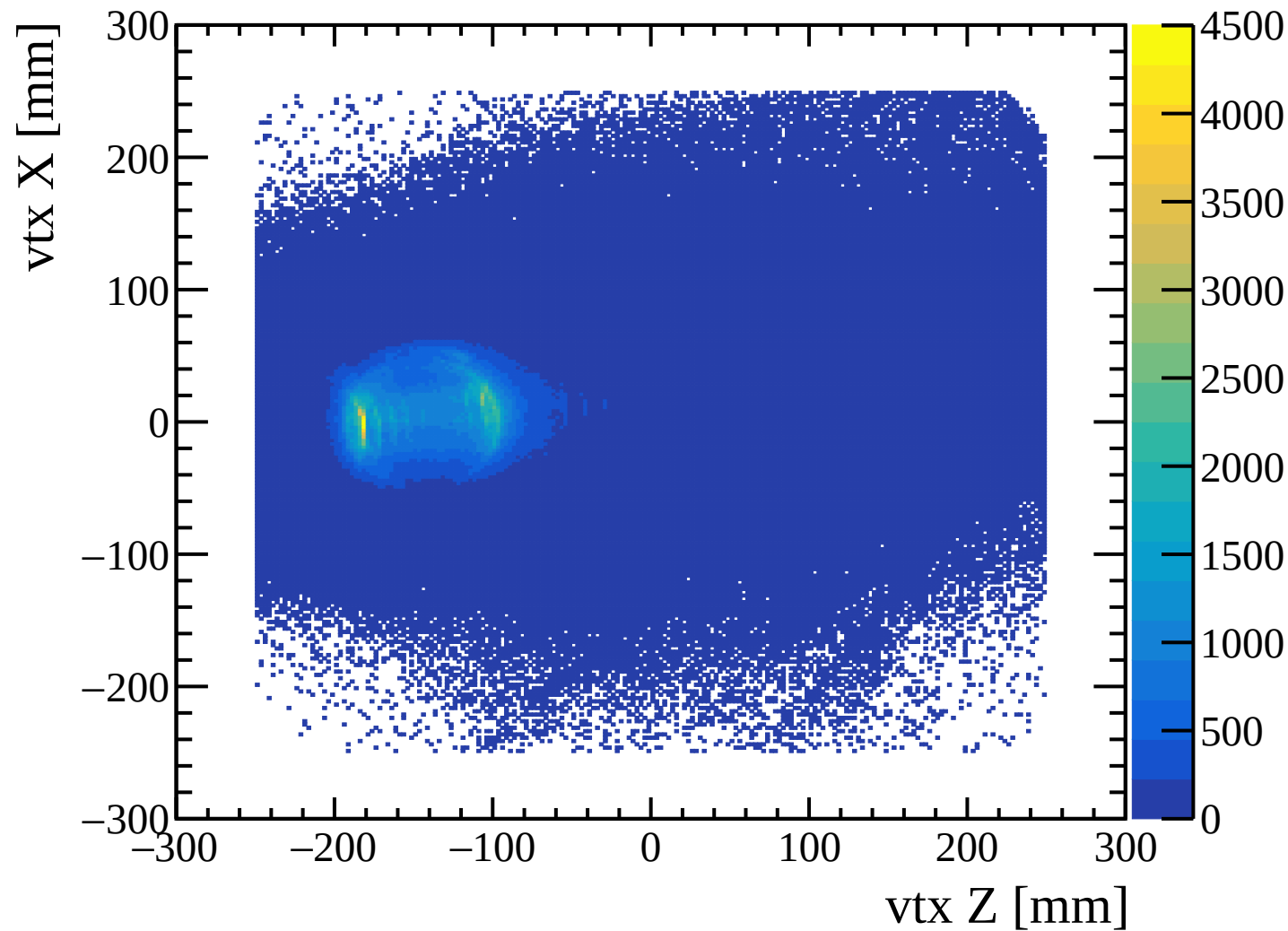
track χ^2 , tight cut



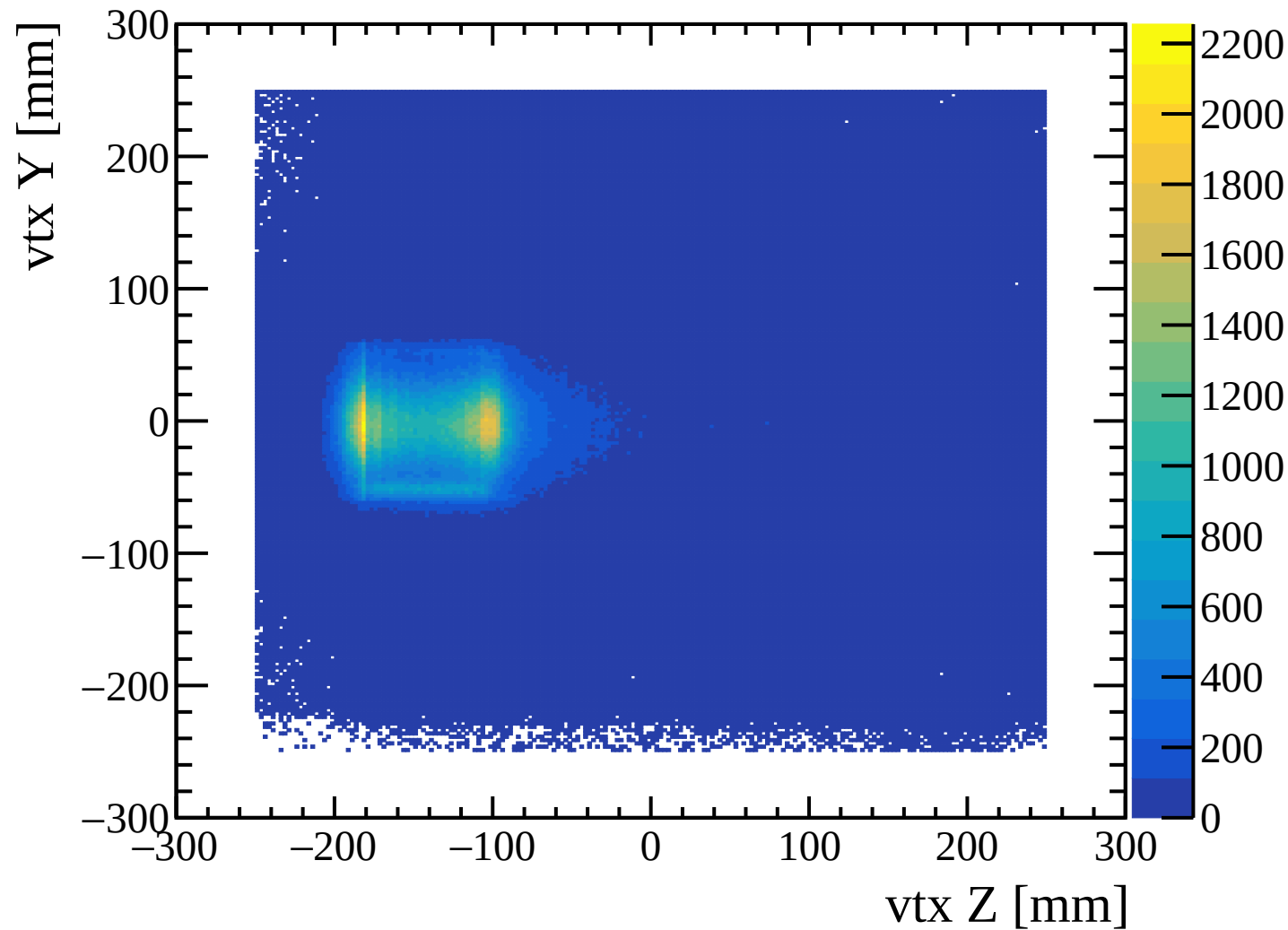
track PID bit pattern



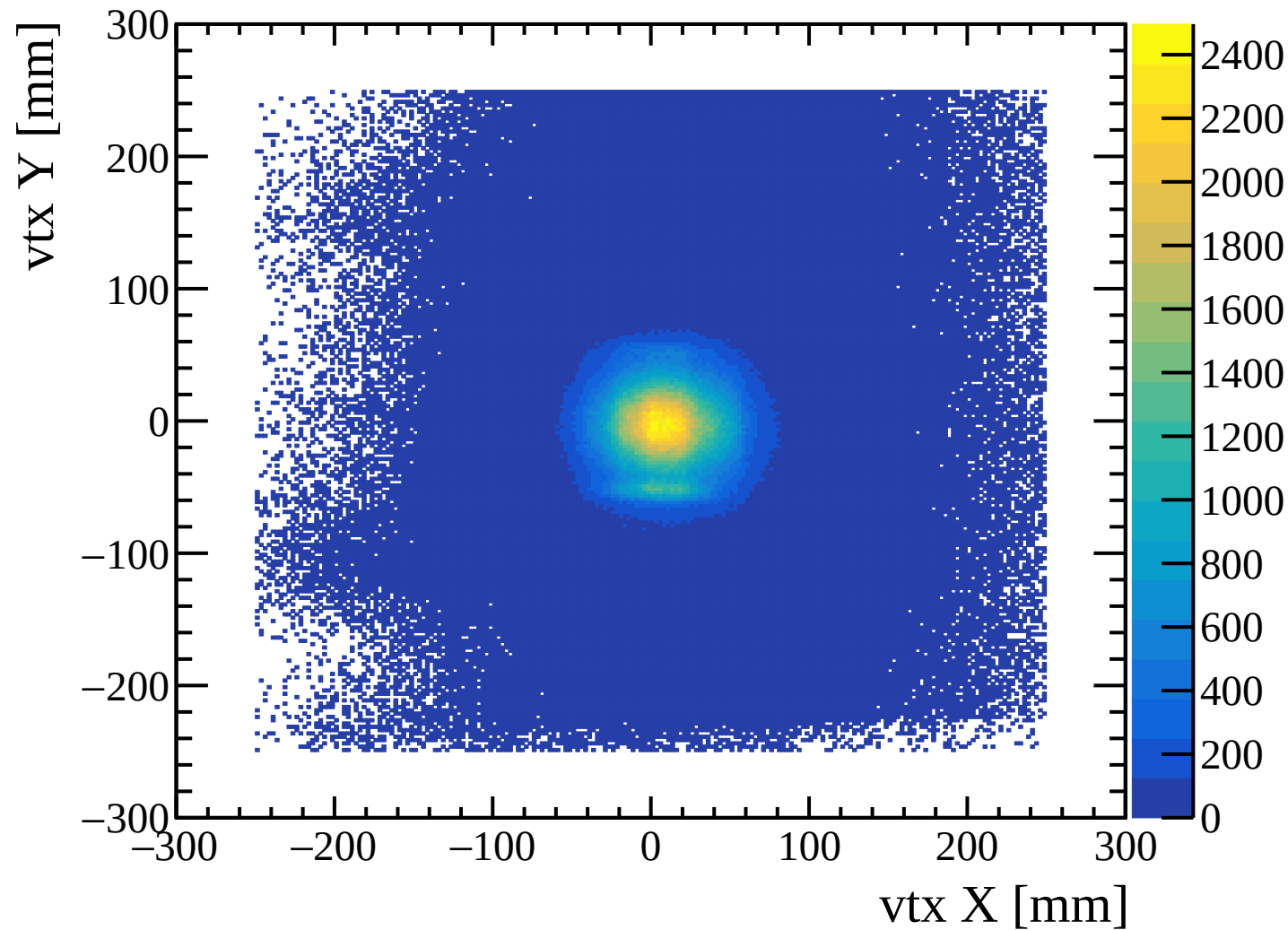
vertex, all $p\pi^-$



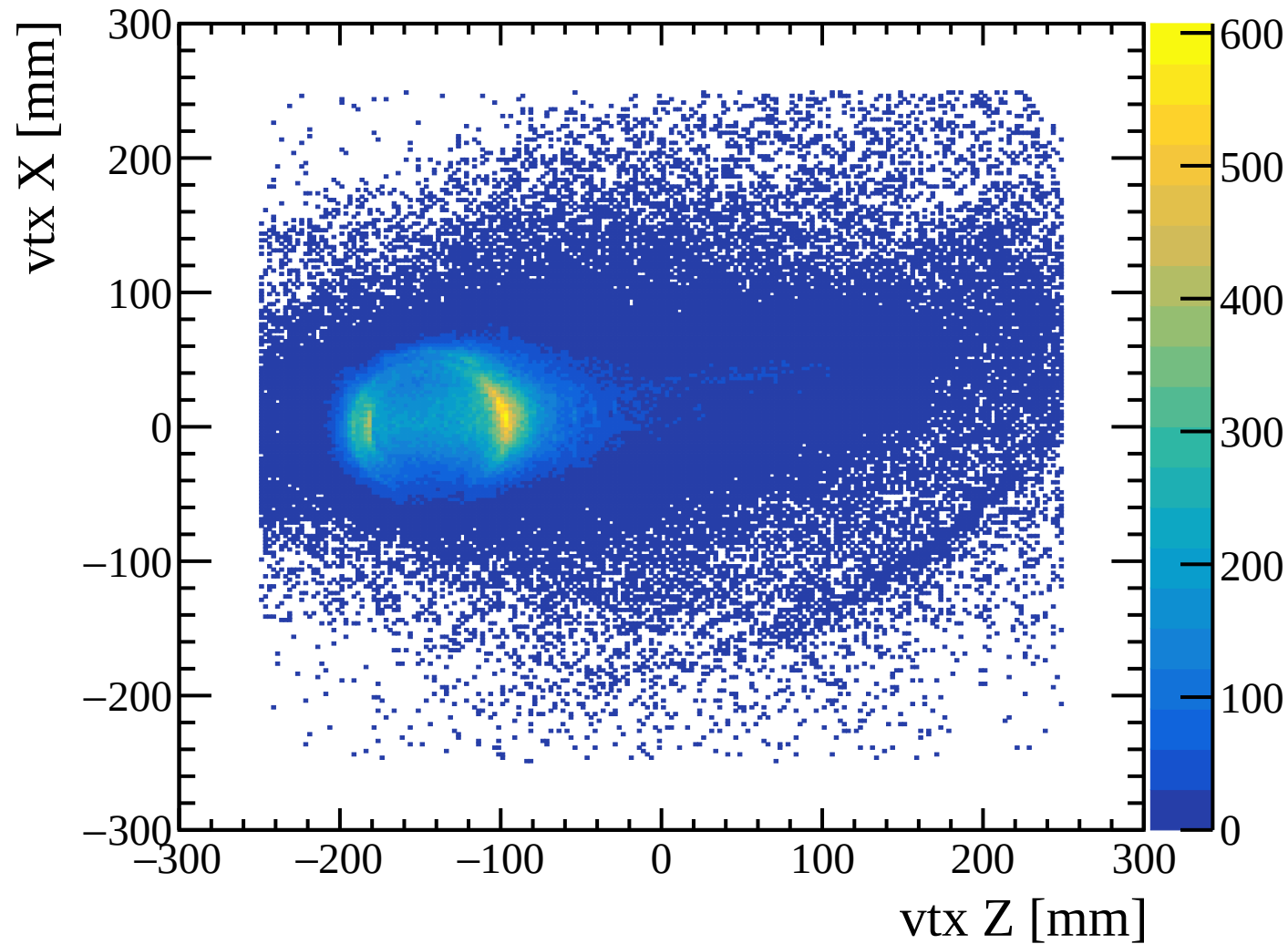
vertex, all $p\pi^-$



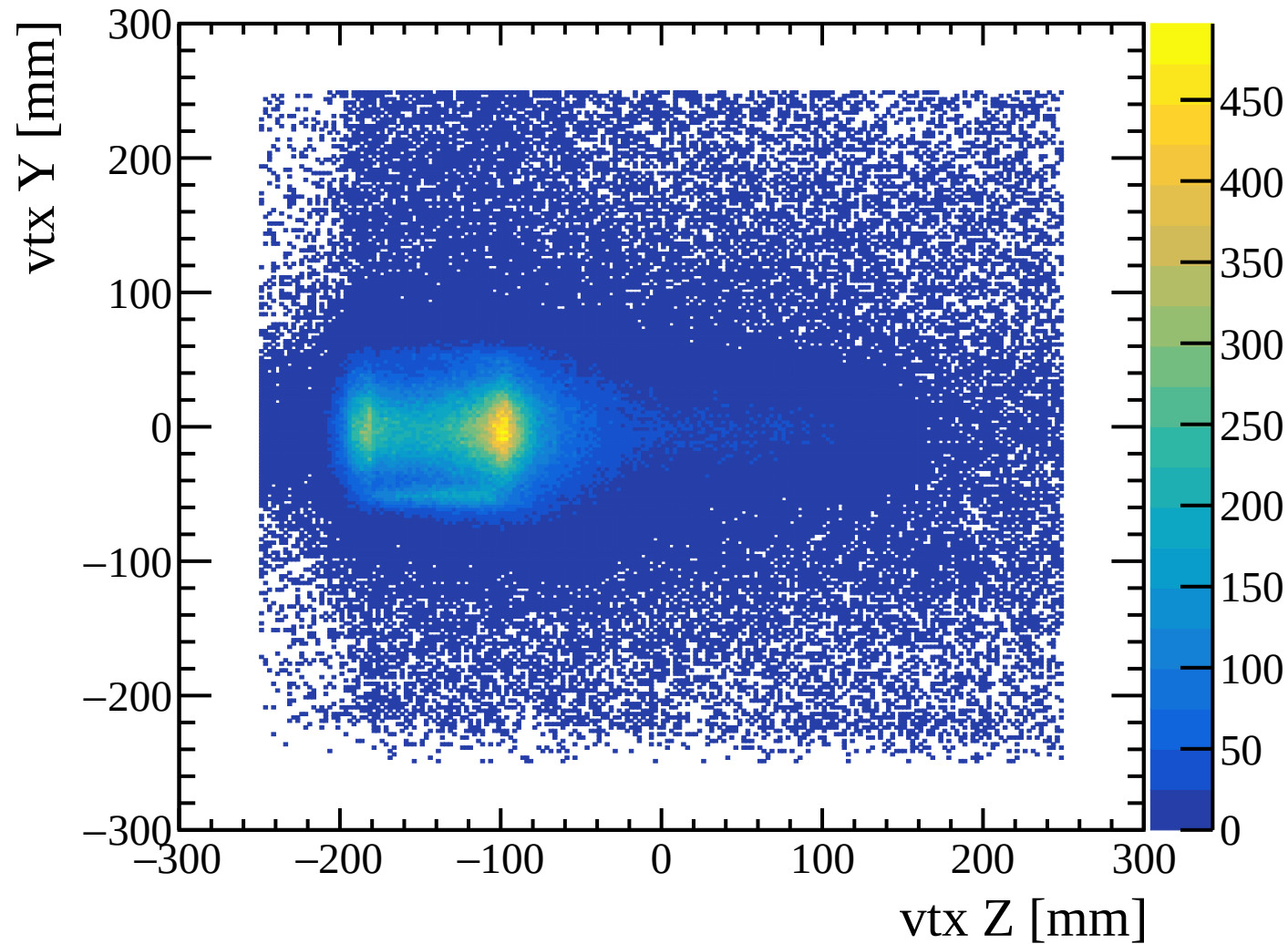
vertex, all $p\pi^-$



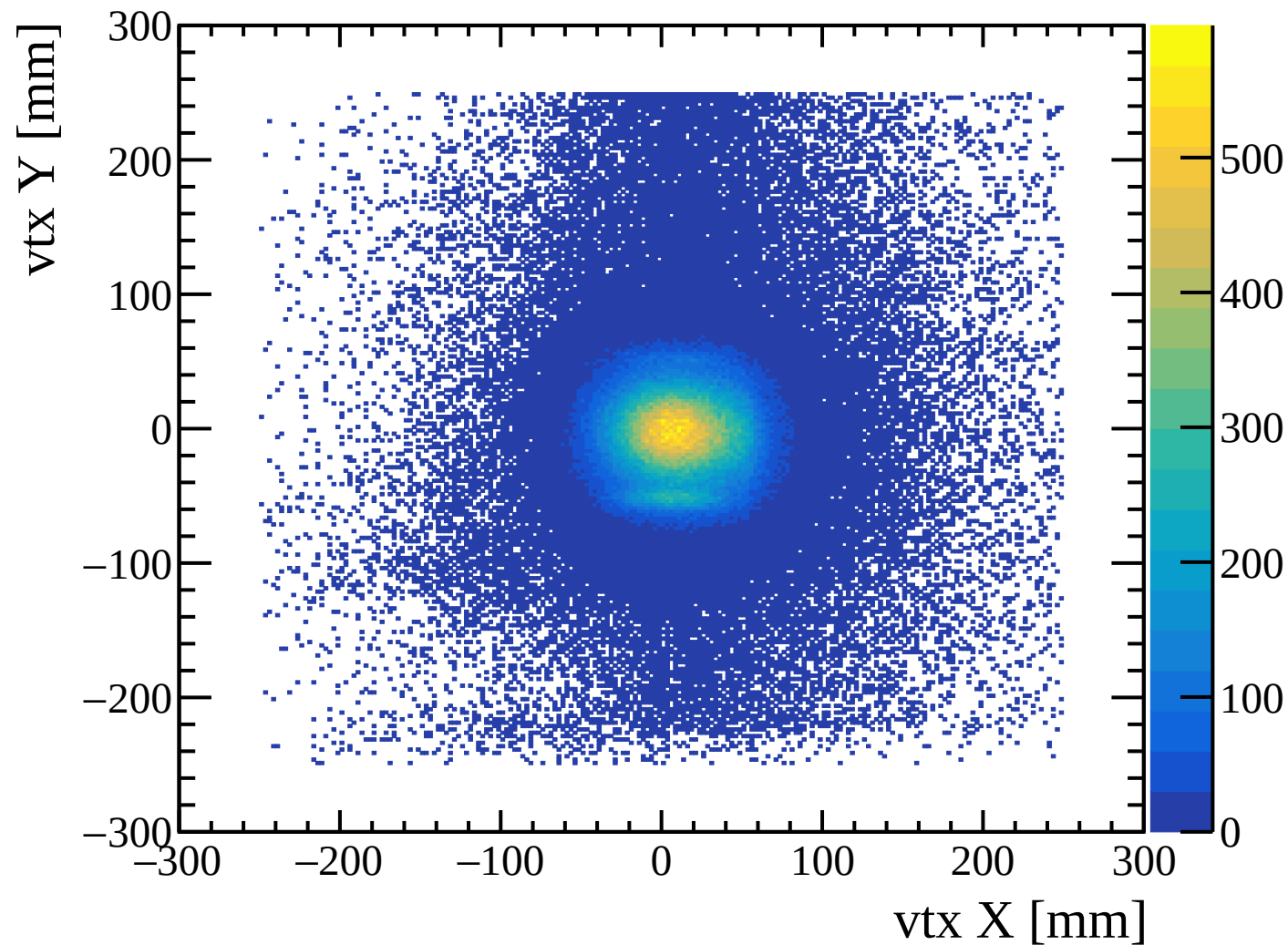
vertex, Lambda peak



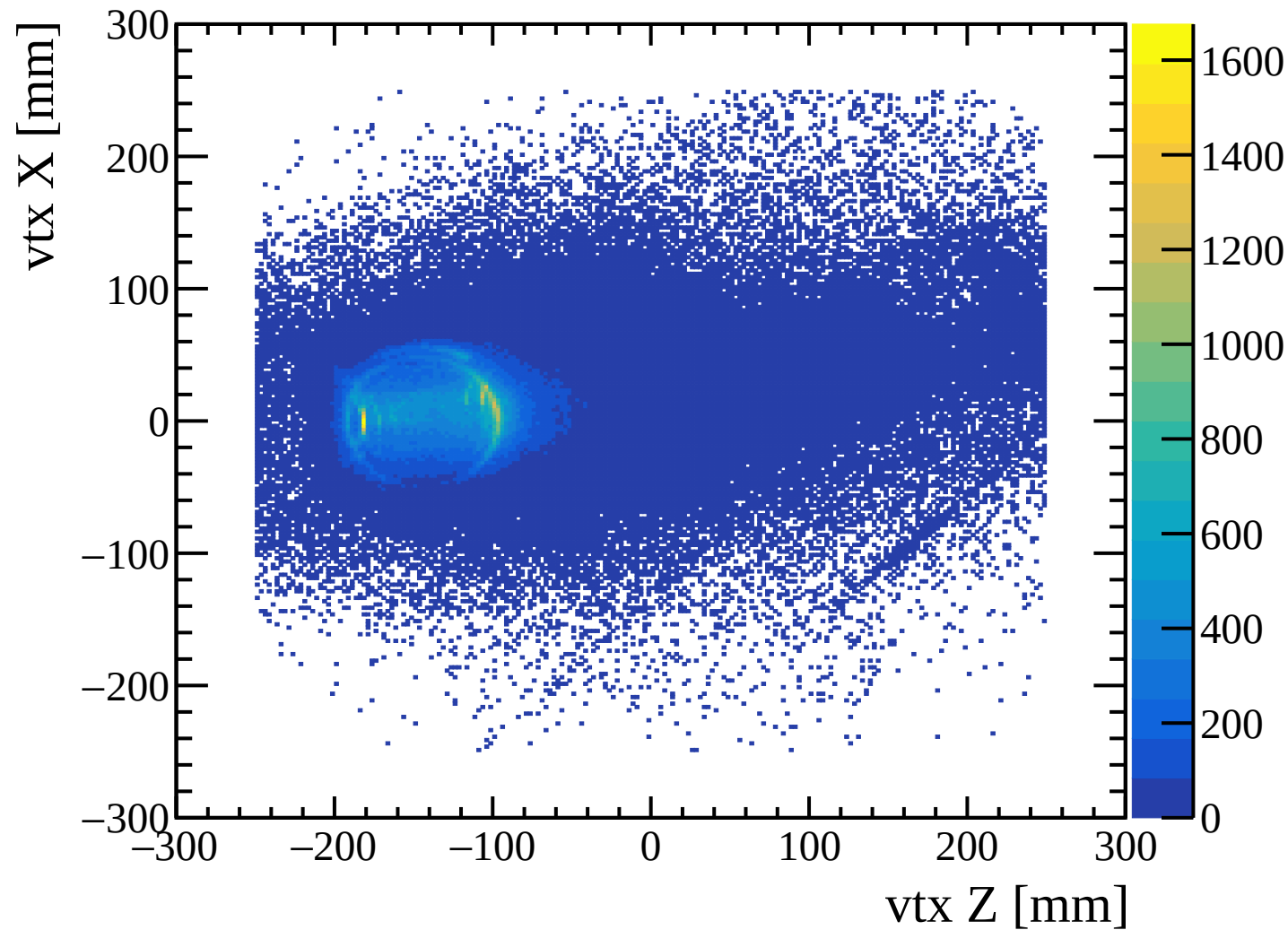
vertex, Lambda peak



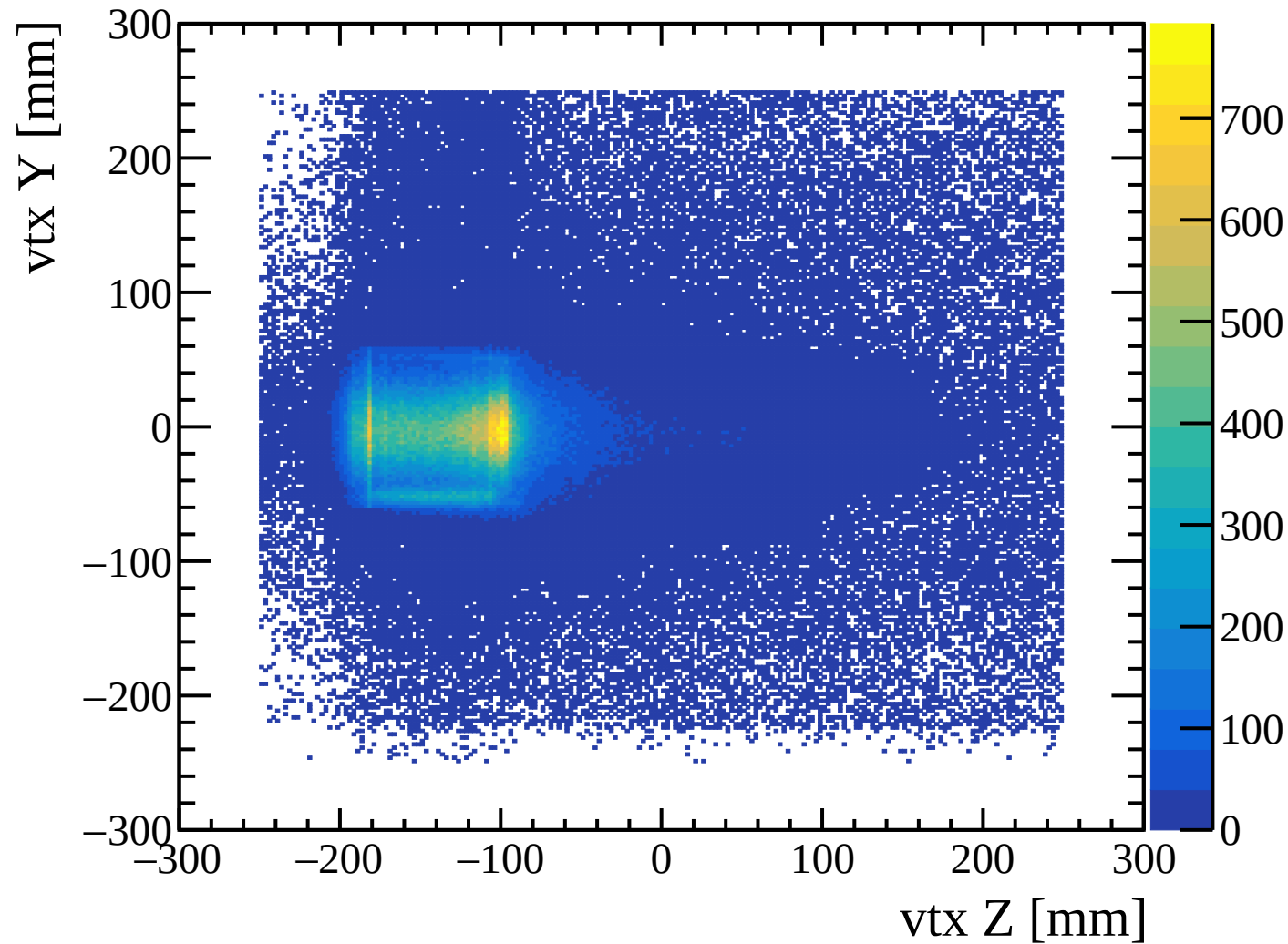
vertex, Lambda peak



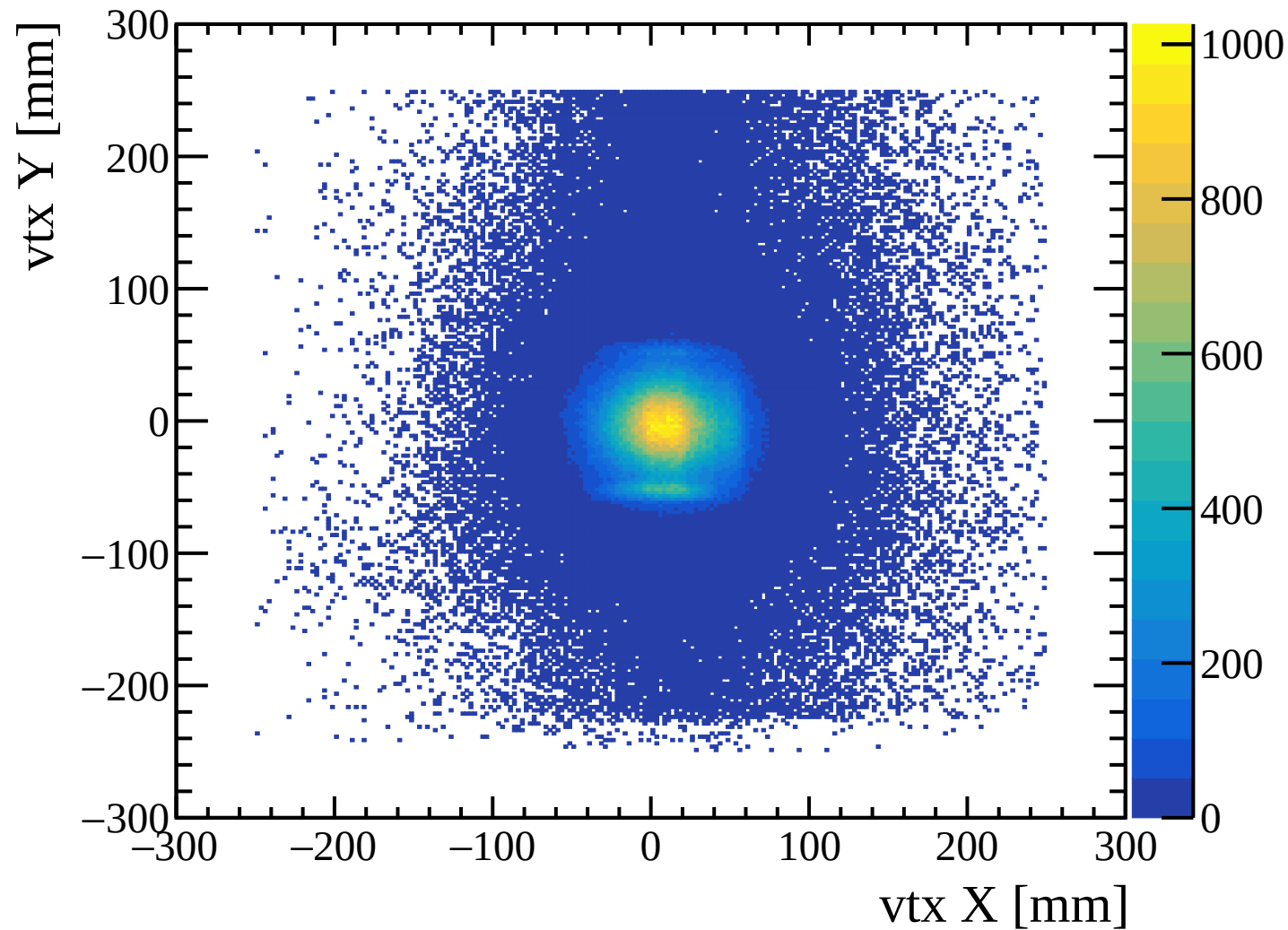
vertex, tight cut



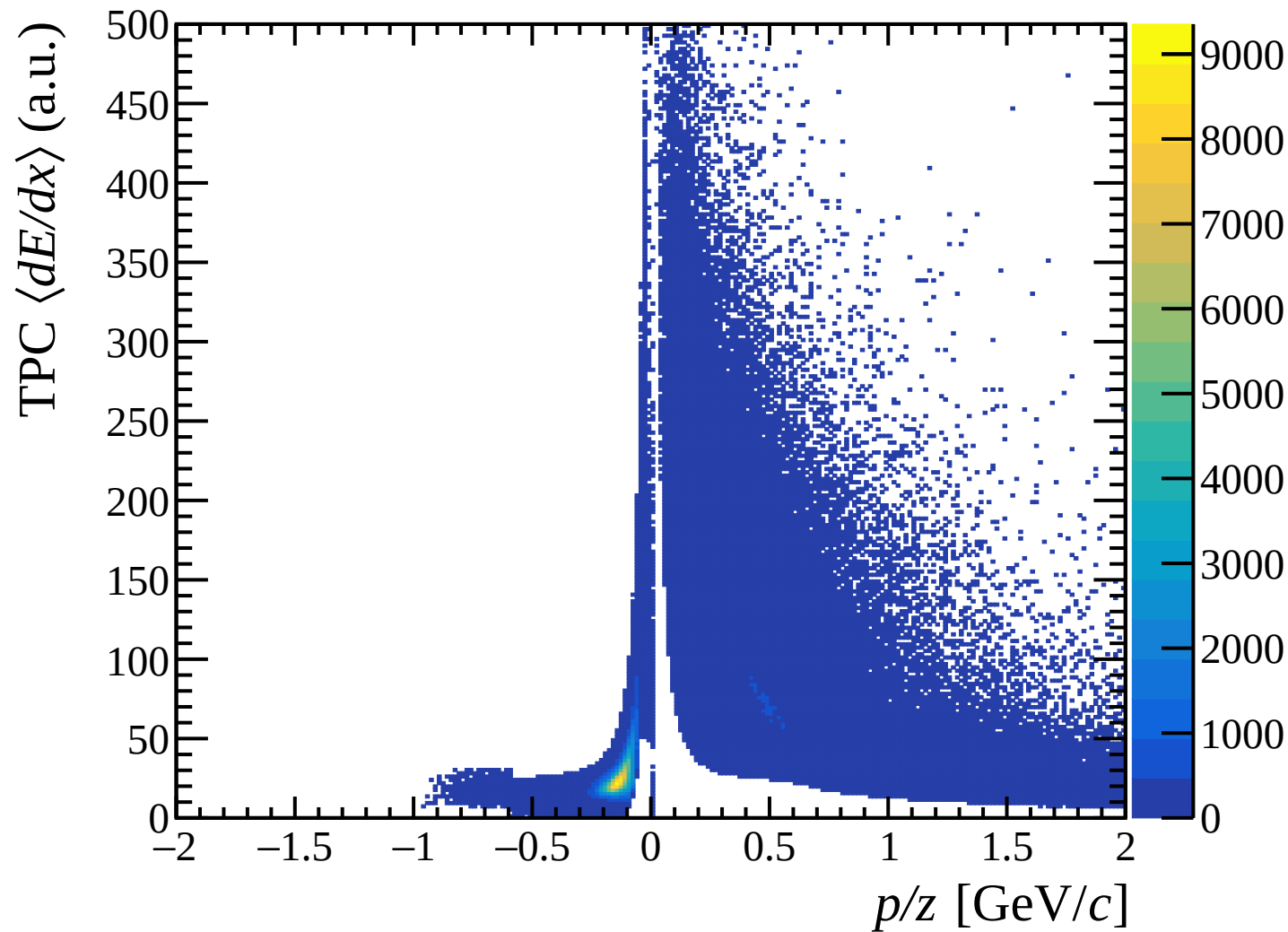
vertex, tight cut



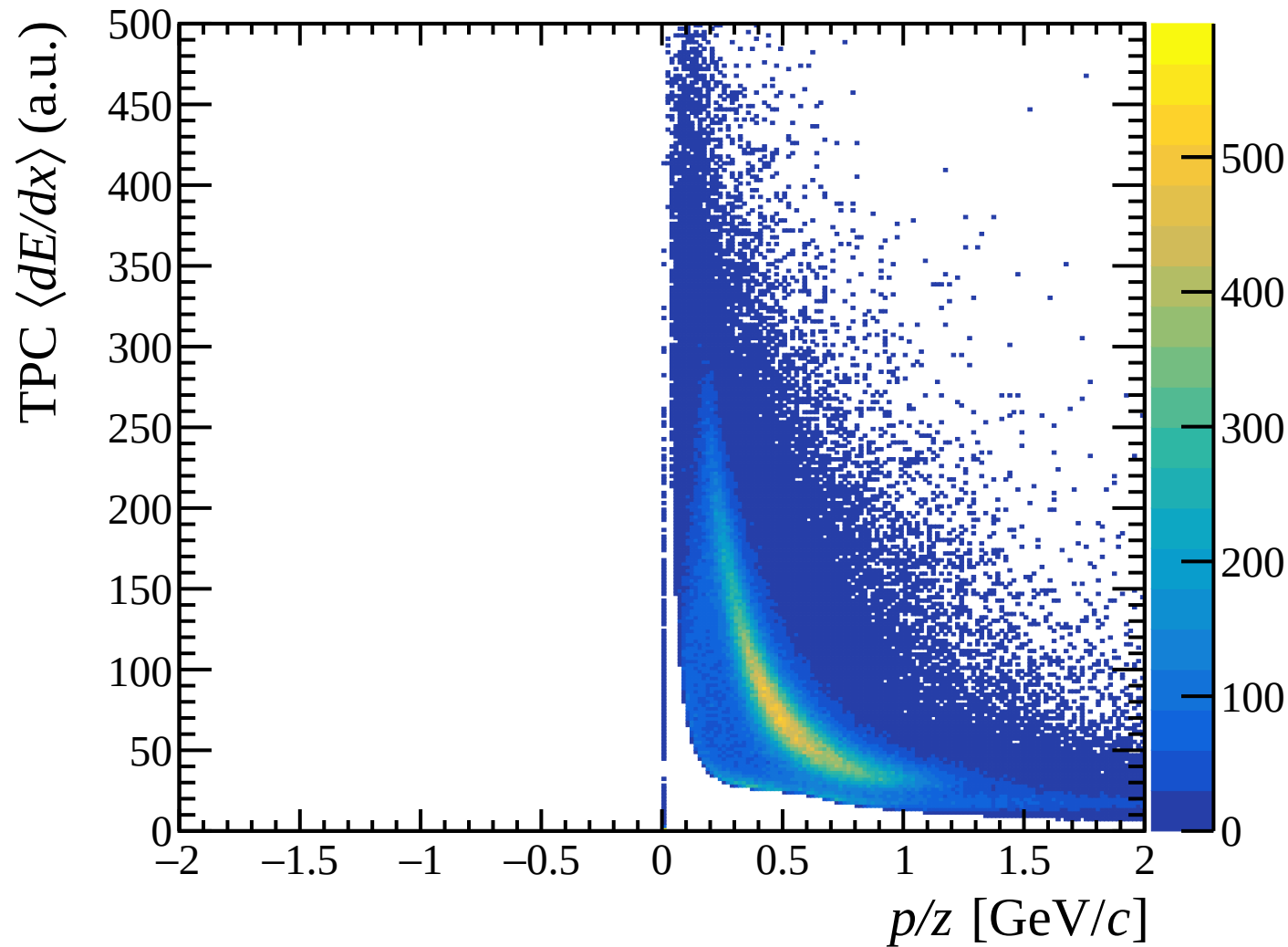
vertex, tight cut



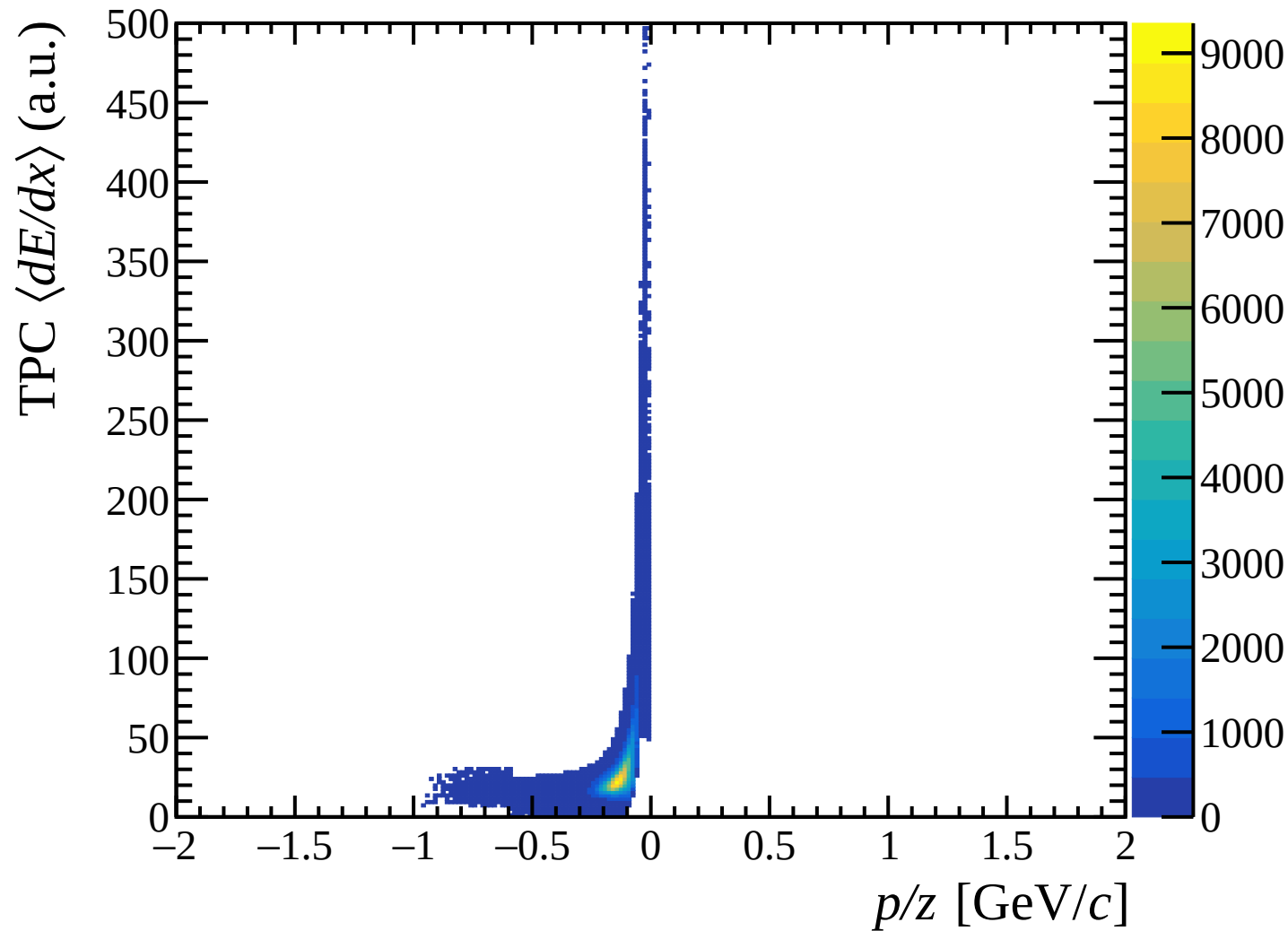
dE/dx vs signed momentum, Lambda peak



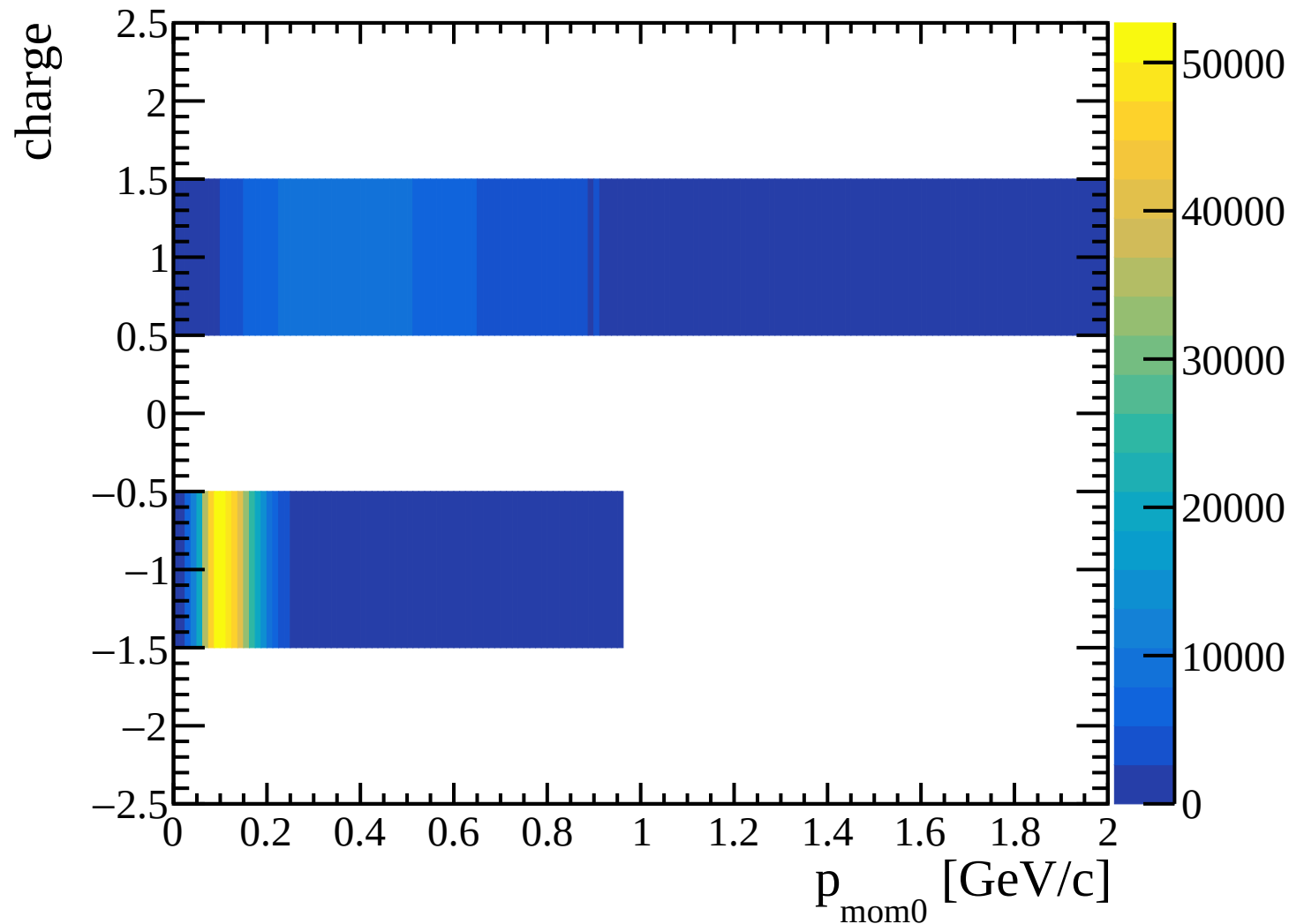
proton dE/dx vs signed momentum, Lambda peak



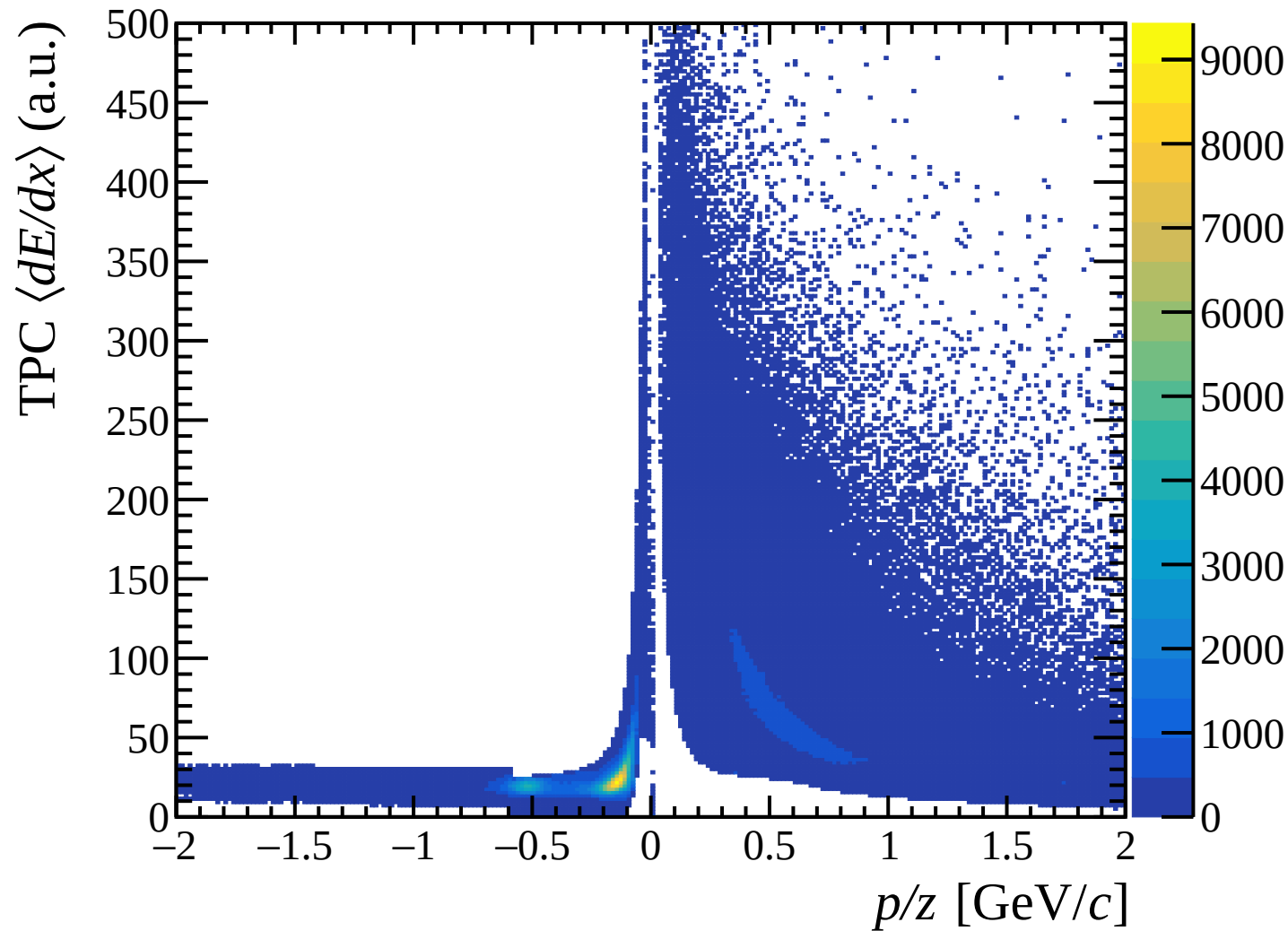
π^- dE/dx vs signed momentum, Lambda peak



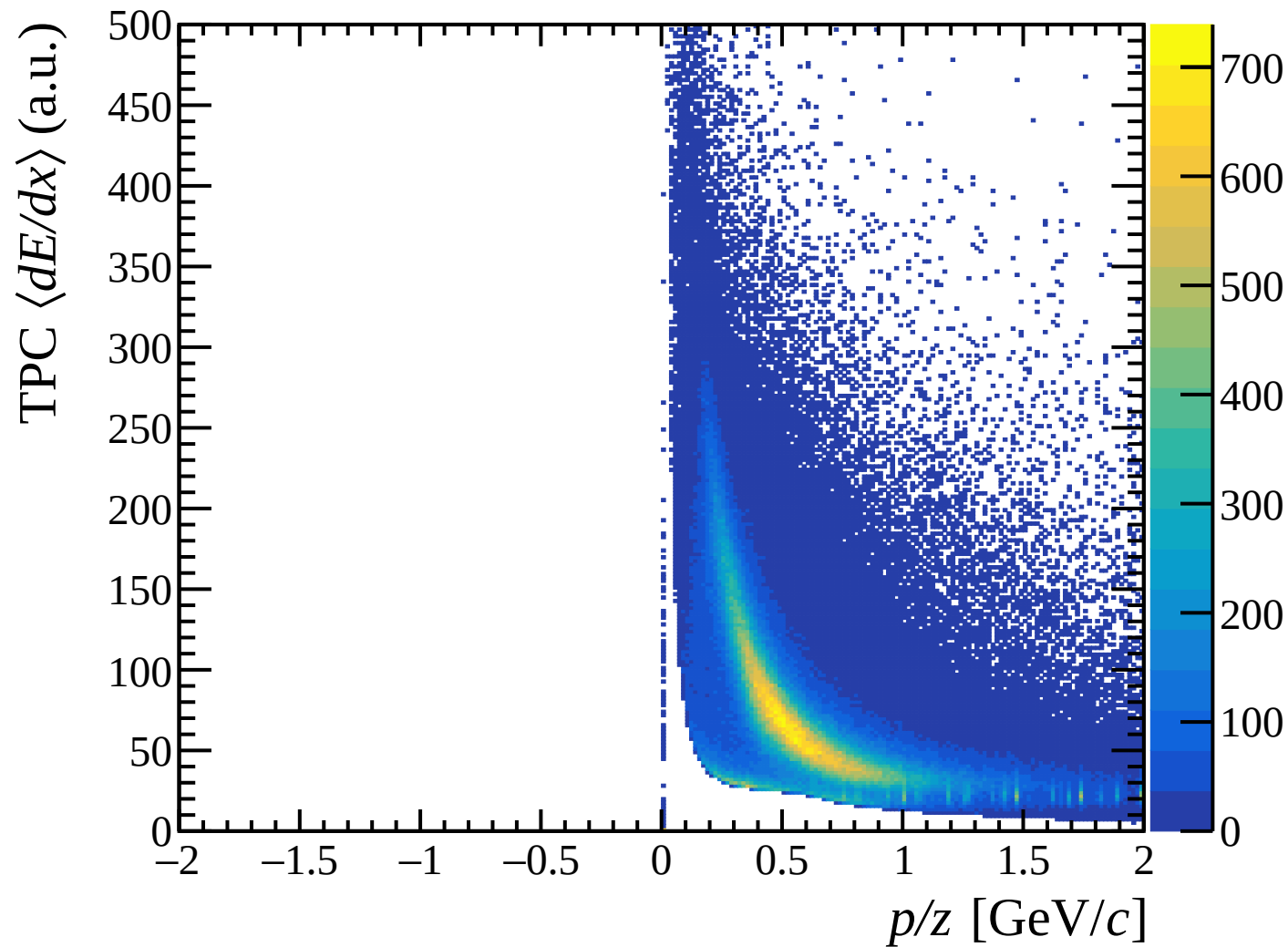
charge vs momentum, Lambda peak



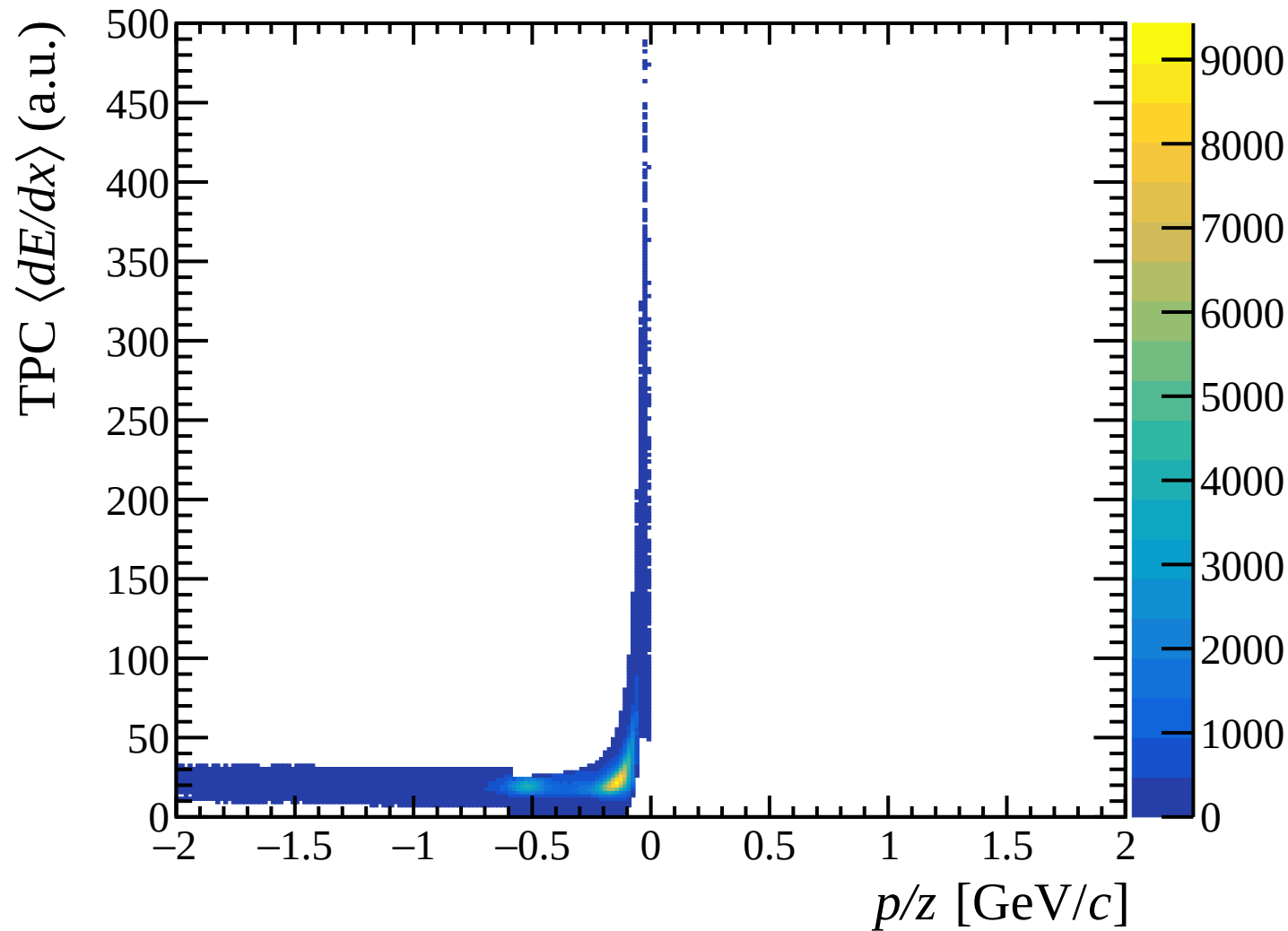
dE/dx vs signed momentum, tight cut



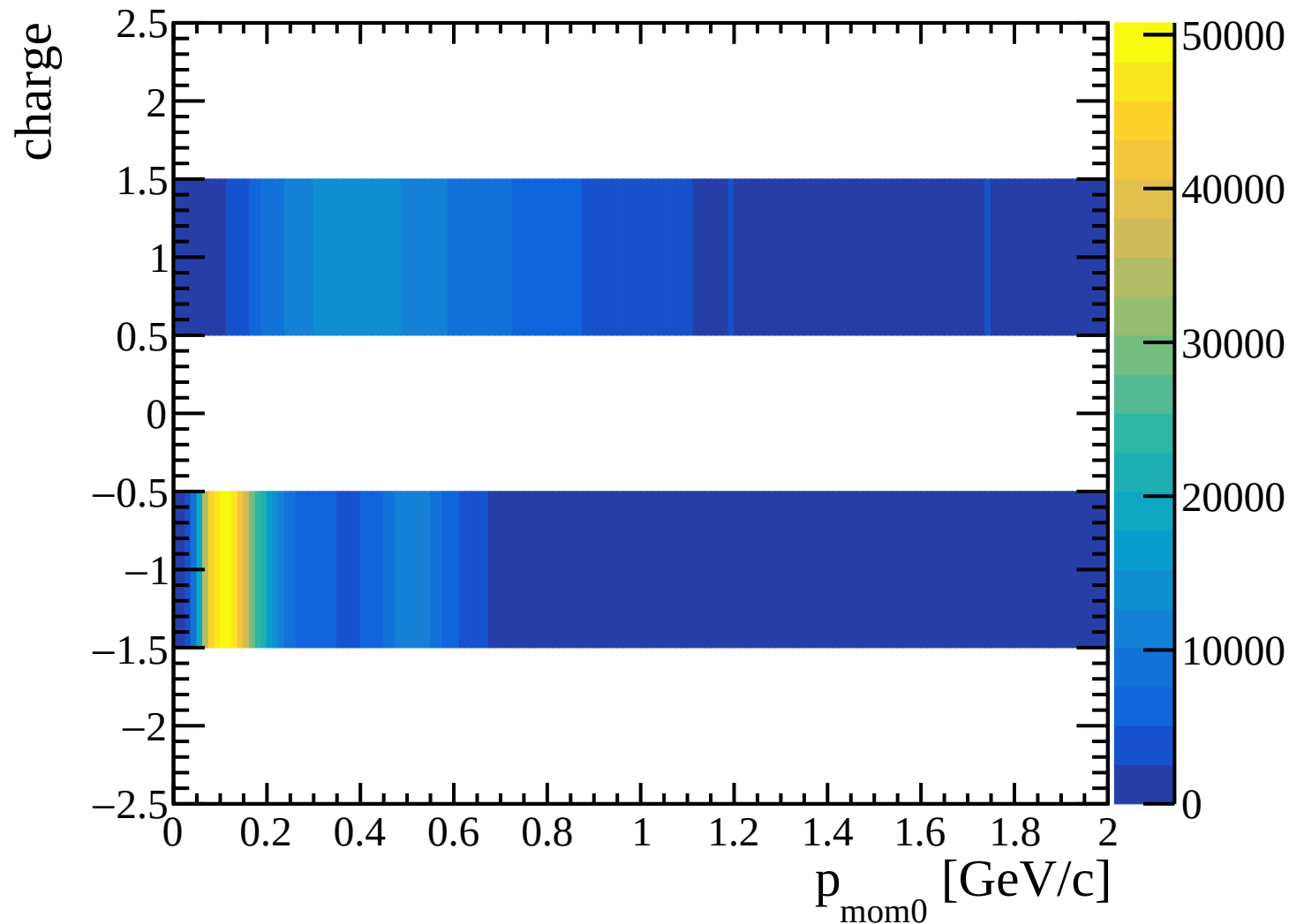
proton dE/dx vs signed momentum, tight cut



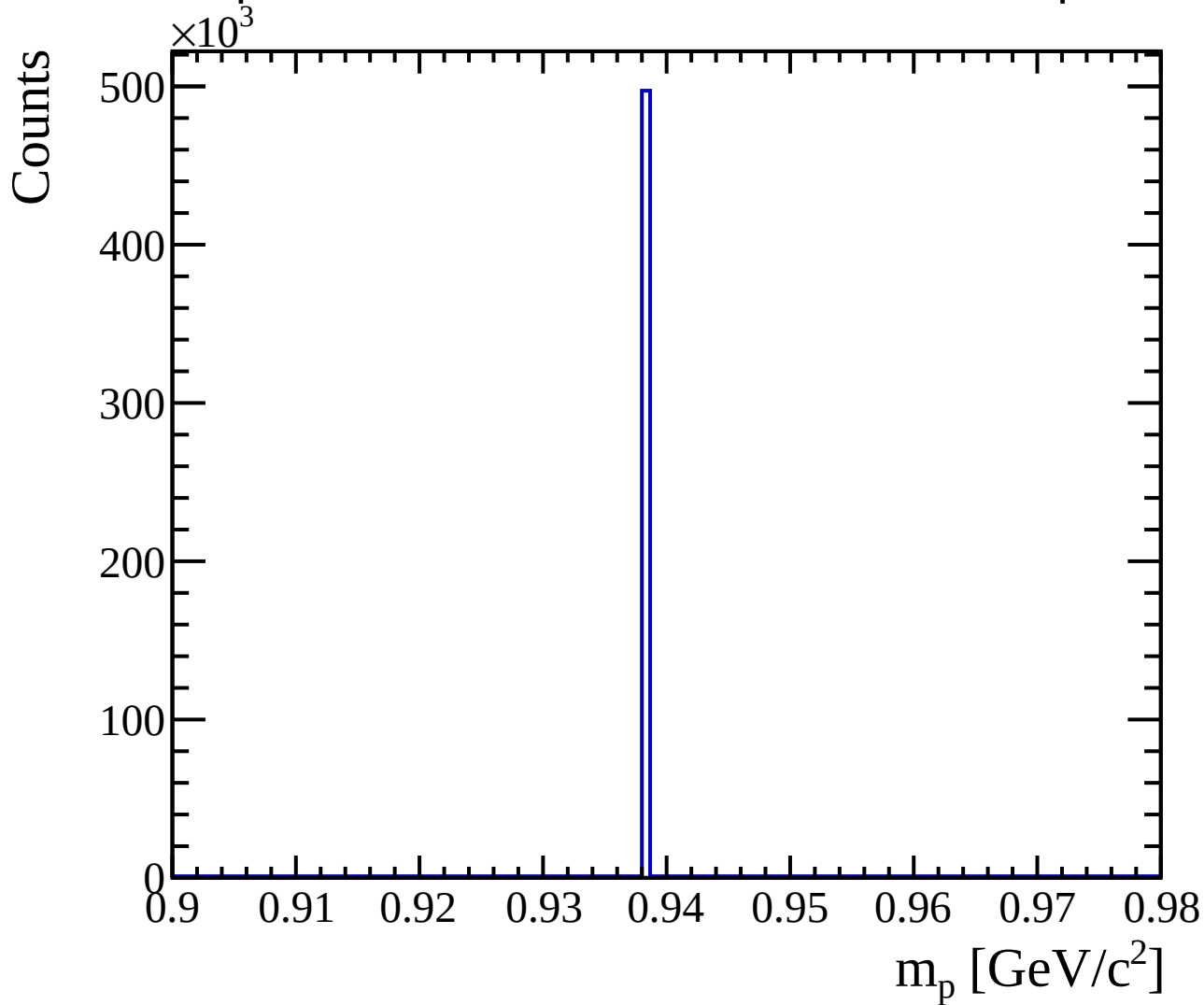
π^- dE/dx vs signed momentum, tight cut



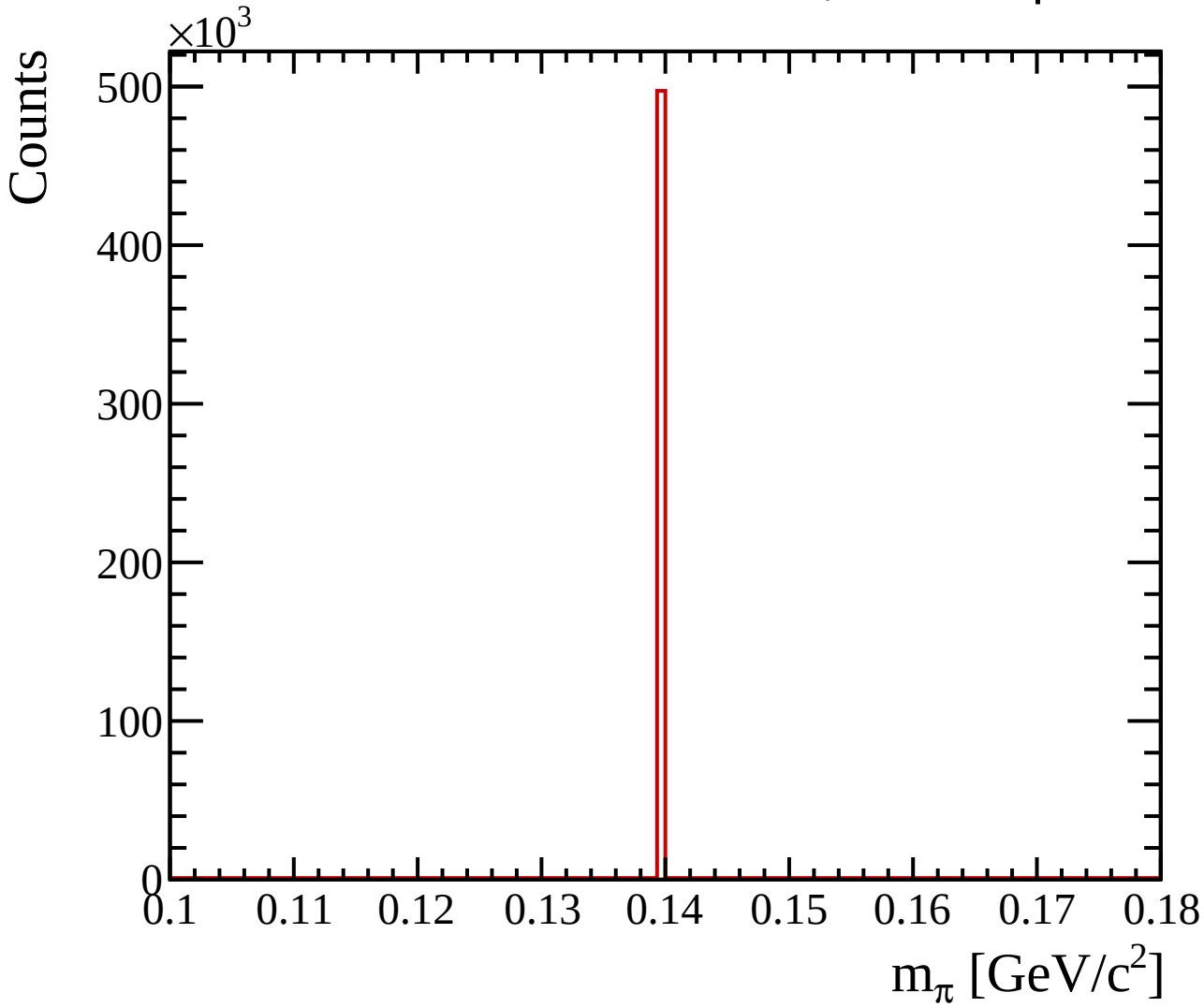
charge vs momentum, tight cut



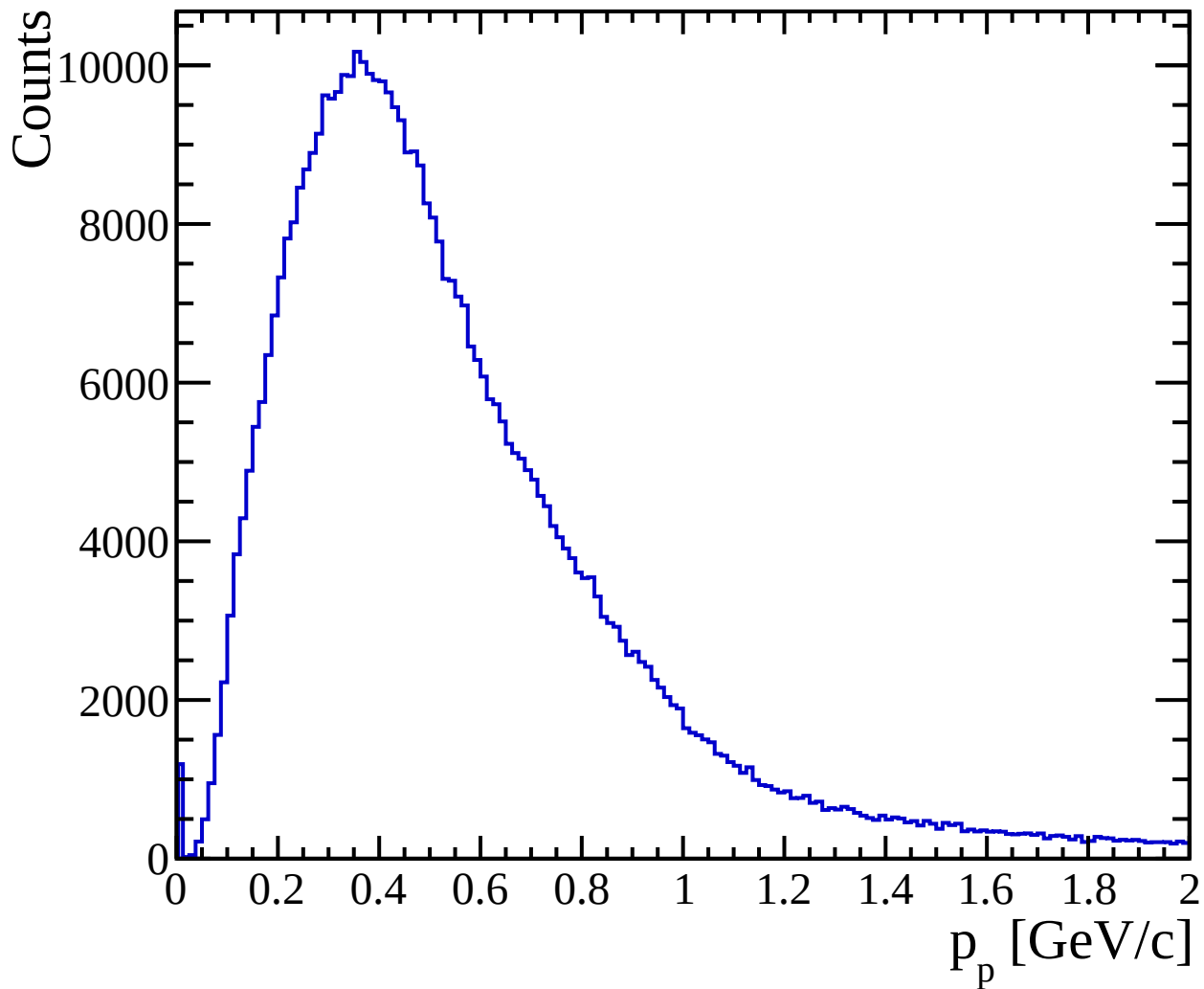
proton candidate invariant mass, Lambda peak



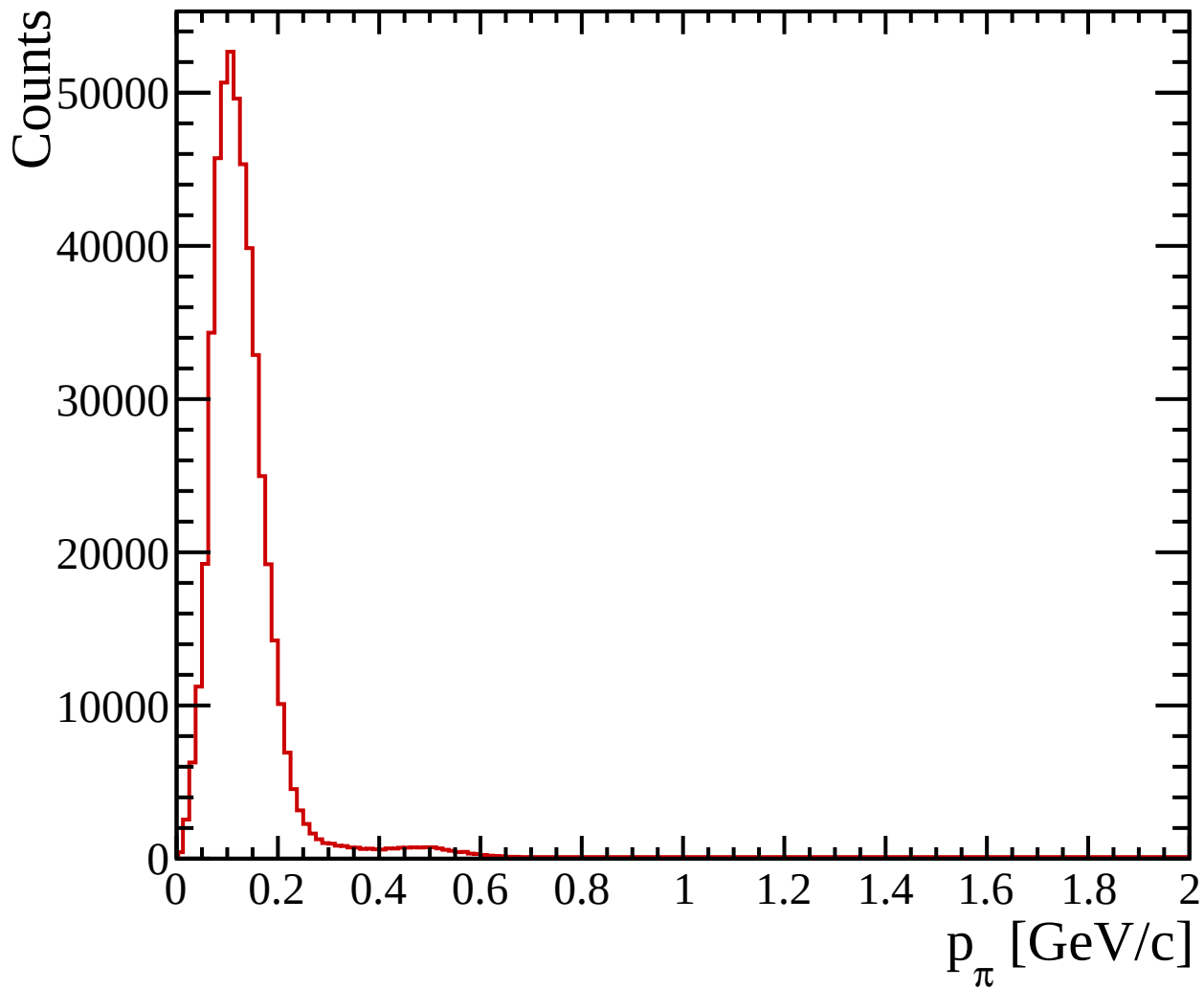
π^- candidate invariant mass, Lambda peak



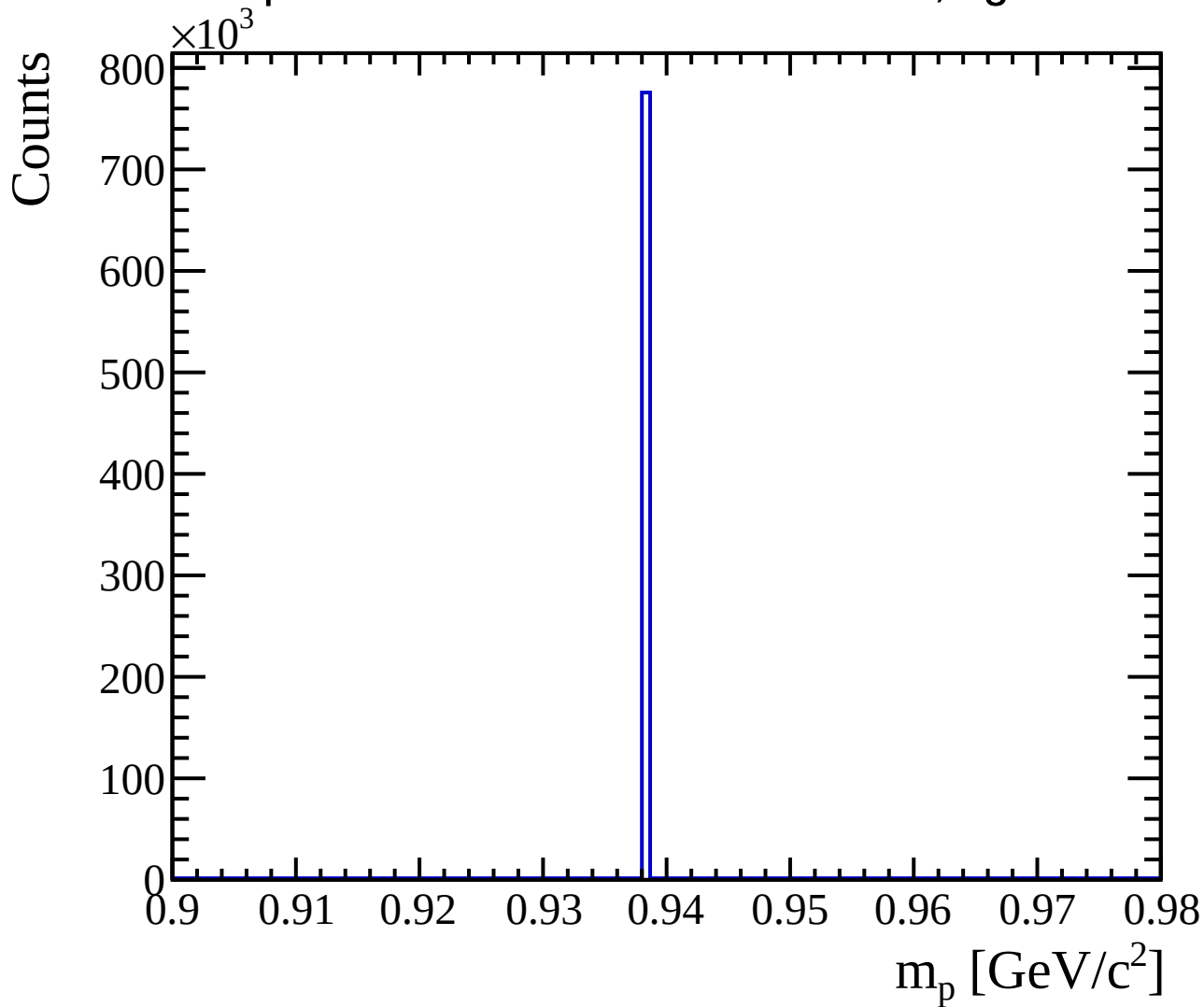
proton momentum at mom0, Lambda peak



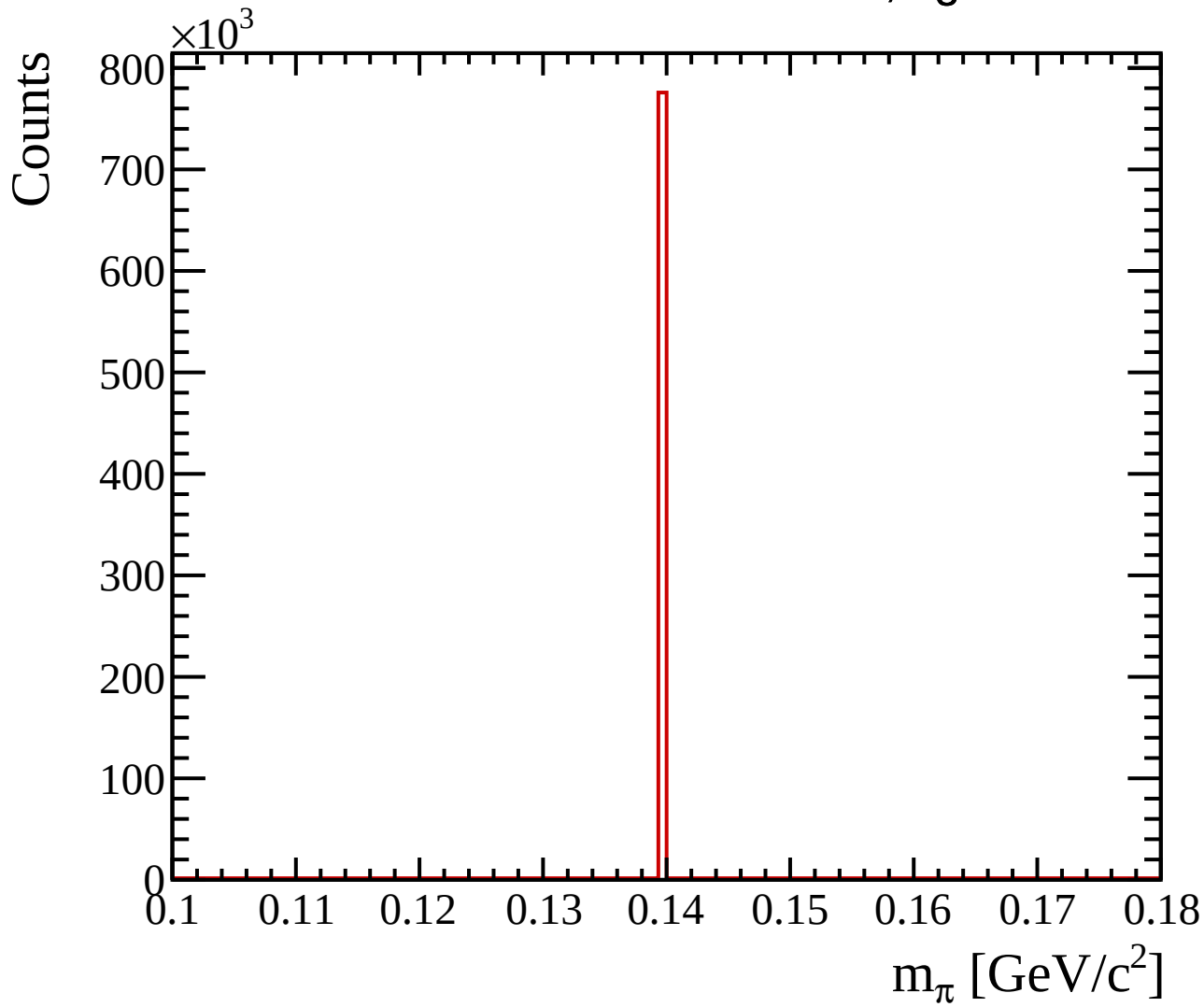
π^- momentum at mom0, Lambda peak



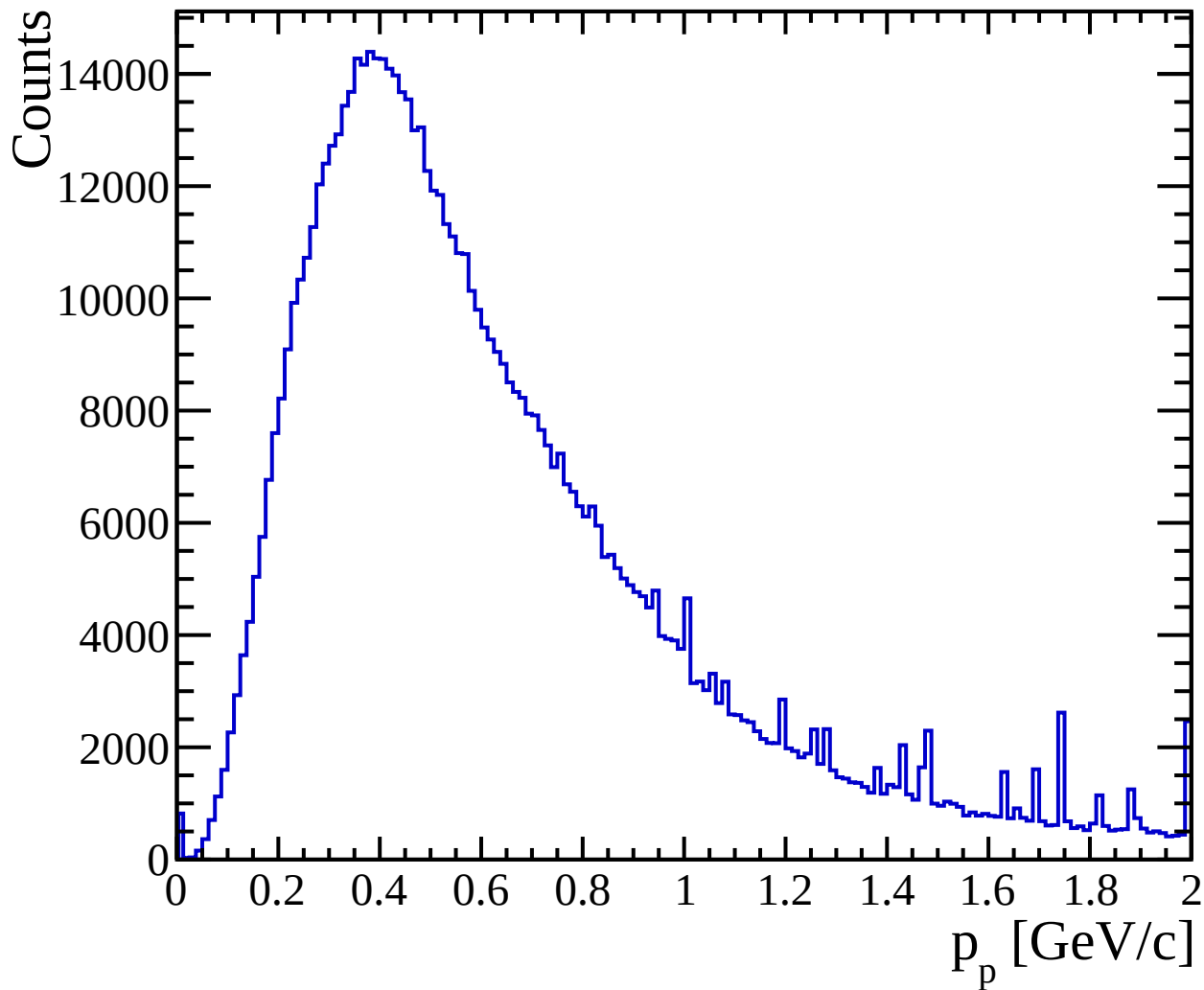
proton candidate invariant mass, tight cut



π^- candidate invariant mass, tight cut



proton momentum at mom0, tight cut



π^- momentum at mom0, tight cut

